

BEACON EDUCATIONAL SERIES

Mathematics for Basic 7

A Comprehensive Question Bank
with Marking Schemes

Curriculum-Aligned Practice Papers
for Basic 7 (JHS 1) Students and Teachers

Beacon Educational Consult
2026 Edition

Mathematics for Basic 7
First Edition, 2026

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This book is designed to support the Ghana Education Service (GES) curriculum for Junior High School. It is intended as a supplementary resource for teachers and students.

The questions in this book are original practice materials created for educational purposes. They are not official BECE or GES examination questions.

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PREFACE

This book has been developed to support Junior High School Basic 7 (JHS 1) students and teachers in their study of Mathematics, as prescribed by the Ghana Education Service (GES) and the National Council for Curriculum and Assessment (NaCCA).

The book contains a comprehensive collection of practice questions organised according to the curriculum's strand and sub-strand structure. Each paper includes both multiple-choice questions (Section A) and structured theory questions (Section B), reflecting the format of BECE examinations.

Key Features

- Curriculum-Aligned Content — All questions mapped to specific content standards.
- Comprehensive Coverage — Topic-based papers covering all major strands.
- Detailed Marking Schemes — Complete answer keys with method marks and marking guidance.
- Professional Diagrams — Clear, scientifically accurate illustrations.
- Cognitive Demand — Questions spanning all levels of Bloom's taxonomy.

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ANSWER KEY

Chapter 1: Ch10

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. The mathematical instruments used to construct perpendiculars in class are
 - A. a protractor and a calculator
 - B. a pair of compasses and a ruler
 - C. a measuring tape and a balance
 - D. a clock and a set square
2. Two lines are perpendicular if the angle between them is
 - A. 90°
 - B. 45°
 - C. 180°
 - D. 60°
3. To construct an exact copy of a line segment PQ, we use
 - A. a protractor only
 - B. a balance
 - C. a pair of compasses and a ruler
 - D. a calculator
4. To drop a perpendicular means to
 - A. draw a line parallel to the given line
 - B. erect a perpendicular at an end point of the line
 - C. extend the given line
 - D. draw a perpendicular from a point outside the line to meet the line
5. The symbol used to show that two lines are perpendicular is
 - A. $\parallel$
 - B. $\perp$
 - C. $\cdot$
 - D. $\angle$
6. A perpendicular PS is erected at a point P on the line segment AB. The angle APS measures
 - A. 45°
 - B. 180°
 - C. 60°
 - D. 90°

- 7.** To copy a line segment PQ onto a ray starting from a point R, we
- A. open the compasses to the length PQ and draw an arc from R to cut the ray
 - B. use a protractor to mark the length
 - C. estimate the middle of PQ
 - D. draw a line parallel to PQ
- 8.** A line perpendicular to a horizontal line is
- A. also horizontal
 - B. at 45° to it
 - C. vertical
 - D. curved
- 9.** The perpendicular bisector of a line segment cuts the segment at
- A. one end
 - B. its midpoint
 - C. any point
 - D. one quarter of its length
- 10.** A line segment $AB = 8$ cm is bisected perpendicularly. Each half measures
- A. 16 cm
 - B. 2 cm
 - C. 8 cm
 - D. 4 cm
- 11.** When constructing the perpendicular bisector of AB, the compasses should be opened to a width
- A. more than half of AB
 - B. exactly half of AB
 - C. less than half of AB
 - D. exactly equal to AB
- 12.** A line segment $CD = 5.5$ cm is bisected perpendicularly. Each half measures
- A. 2.25 cm
 - B. 2.75 cm
 - C. 5.5 cm
 - D. 11 cm
- 13.** In constructing the perpendicular bisector of a segment, the arcs are drawn
- A. from one end point only
 - B. from the midpoint
 - C. from a point outside the line
 - D. from both end points with equal radii

- 14.** Any point on the perpendicular bisector of AB is
- A. equidistant from A and B
 - B. always nearer to A than to B
 - C. twice as far from B as from A
 - D. always at the midpoint of AB
- 15.** A line segment is 12.4 cm long. The distance from one end to the point where its perpendicular bisector meets it is
- A. 12.4 cm
 - B. 24.8 cm
 - C. 6.2 cm
 - D. 3.1 cm
- 16.** When constructing a perpendicular bisector, the two arcs cross
- A. on the line segment
 - B. on either side of the line segment
 - C. only above the line
 - D. at the end points
- 17.** An angle bisector divides an angle into
- A. two unequal parts
 - B. three equal parts
 - C. two parts of 90° each
 - D. two equal parts
- 18.** Bisecting an angle of 90° gives two angles of
- A. 45°
 - B. 30°
 - C. 60°
 - D. 22.5°
- 19.** To construct 60° with a pair of compasses, we essentially construct
- A. a square
 - B. a right angle and halve it
 - C. an equilateral triangle
 - D. a perpendicular bisector
- 20.** Bisecting an angle of 60° gives
- A. 30°
 - B. 15°
 - C. 45°
 - D. 20°

- 21.** Bisecting an angle of 30° gives
- A. 20°
 - B. 10°
 - C. 15°
 - D. 45°
- 22.** An angle of 75° can be constructed by
- A. bisecting the angle between 60° and 90°
 - B. bisecting a right angle
 - C. bisecting 60°
 - D. bisecting 30°
- 23.** An angle of 45° can be constructed by
- A. bisecting 60°
 - B. bisecting 90°
 - C. bisecting 30°
 - D. copying 50°
- 24.** Construct $\angle PQR = 75^\circ$ and bisect it. Each of the two angles obtained measures
- A. 75°
 - B. 150°
 - C. 30°
 - D. 37.5°
- 25.** To construct an angle of 15° , we bisect an angle of
- A. 45°
 - B. 60°
 - C. 30°
 - D. 90°
- 26.** When copying an angle with compasses, what is transferred from the original angle to the new line?
- A. the protractor reading
 - B. the area of the angle
 - C. only the lengths of the arms
 - D. an arc and its chord
- 27.** Bisecting $\angle ABC = 120^\circ$ gives two angles of
- A. 240°
 - B. 60°
 - C. 90°
 - D. 30°

- 28.** Construct $\angle XYZ = 60^\circ$ and bisect it, then bisect one of the angles obtained again. The smallest angle obtained is
- A. 30°
 - B. 20°
 - C. 15°
 - D. 45°
- 29.** Which of the following angles can ALL be constructed using a pair of compasses and a ruler only?
- A. 30° , 45° , 60° , 75° and 90°
 - B. 10° , 20° and 40°
 - C. 5° , 35° and 85°
 - D. 50° , 70° and 100°
- 30.** After constructing 90° by raising a perpendicular at a point on a line, the angle is verified using
- A. a ruler
 - B. the compasses
 - C. a balance
 - D. a protractor
- 31.** In the construction of 45° , the line that bisects the right angle is called the
- A. angle bisector
 - B. perpendicular bisector
 - C. hypotenuse
 - D. radius
- 32.** Construct $\angle AOB = 60^\circ$ and draw its bisector OC. The angles AOC and COB each measure
- A. 60°
 - B. 45°
 - C. 30°
 - D. 120°
- 33.** Which of the following best illustrates perpendicular lines in the environment?
- A. the curved rim of a bowl
 - B. the corner of a classroom wall
 - C. the surface of a ball
 - D. the arc of a rainbow
- 34.** A carpenter wants the two sides of a window frame to meet at right angles. The construction idea being applied is
- A. angle bisection
 - B. perpendicular bisector of a line
 - C. copying a line segment

D. perpendicular lines

- 35.** Which everyday situation suggests an angle bisector?
- A. the crease formed when a paper corner is folded so that its two edges match
 - B. a torn edge of paper
 - C. a circular plate
 - D. a straight ruler
- 36.** A rectangular box (crate) has adjacent sides that meet at
- A. acute angles
 - B. right angles
 - C. obtuse angles
 - D. reflex angles
- 37.** A roof frame is symmetric about its central member, which divides the roof angle into two equal parts. The central member acts as
- A. a perpendicular
 - B. a parallel line
 - C. a chord
 - D. an angle bisector
- 38.** Estimating the size of the angle opened by a pair of scissors is an example of
- A. estimating angles in everyday tools
 - B. constructing 60°
 - C. bisecting a line segment
 - D. measuring mass
- 39.** Which of the following pairs of objects meet at right angles?
- A. two parallel railway lines
 - B. the opposite edges of a blackboard
 - C. the floor and the wall of a classroom
 - D. the two long edges of a ruler
- 40.** Constructions such as perpendiculars and bisectors are useful in daily life because they help us to
- A. count money accurately
 - B. design and check structures such as buildings and boxes
 - C. memorise multiplication tables
 - D. weigh objects

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) Perpendiculars and copying line segments.

Question 2

- (a) Perpendicular bisectors of line segments.

Question 3

- (a) Copying and bisecting angles.

Question 4

- (a) Constructing 90° , 45° , 60° and 30° .

Question 5

- (a) Constructing 15° and 75° .

Question 6

- (a) Constructions in the environment.

Chapter 2: Ch1

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Write 7 cm 5 mm in centimetres.
 - A. 7.05 cm
 - B. 75 cm
 - C. 7.5 cm
 - D. 12 cm
2. Write 1 m 75 cm in metres.
 - A. 1.075 m
 - B. 1.75 m
 - C. 17.5 m
 - D. 76 m
3. Write 2 km 300 m in metres.
 - A. 500 m
 - B. 230 m
 - C. 23 000 m
 - D. 2300 m
4. The sides of a triangle are 3 cm, 4 cm 5 mm and 5 cm 5 mm. Its perimeter is
 - A. 13 cm
 - B. 12 cm
 - C. 13.5 cm
 - D. 15 cm
5. A rectangle has length 2 m 40 cm and width 1 m 60 cm. Its perimeter is
 - A. 4 m
 - B. 16 m
 - C. 8 m
 - D. 6 m
6. Each side of a square measures 5 cm 4 mm. The perimeter of the square is
 - A. 21.6 cm
 - B. 9.4 cm
 - C. 20.16 cm
 - D. 22 cm

- 7.** Write 4500 m in kilometres.
- A. 45 km
 - B. 0.45 km
 - C. 450 km
 - D. 4.5 km
- 8.** A rectangular field measures 120 m by 80 m. The distance round the field is
- A. 200 m
 - B. 400 m
 - C. 9600 m
 - D. 320 m
- 9.** The perimeter of an equilateral triangle of side 4 cm 2 mm is
- A. 12.2 cm
 - B. 16.8 cm
 - C. 12.6 cm
 - D. 8.6 cm
- 10.** Find the length of wire needed to fence a square garden of side 25 m 50 cm once round.
- A. 102 m
 - B. 100 m
 - C. 51 m
 - D. 204 m
- 11.** Write 3 m 8 cm in metres.
- A. 3.8 m
 - B. 30.8 m
 - C. 3.18 m
 - D. 3.08 m
- 12.** The four sides of a field measure 1 km 200 m, 800 m, 1 km and 1 km. The perimeter of the field, in kilometres, is
- A. 3 km
 - B. 4 km
 - C. 4.2 km
 - D. 40 km
- 13.** The distance round a circle is called the
- A. circumference
 - B. radius
 - C. diameter
 - D. arc

- 14.** The distance from the centre of a circle to any point on the circle is the
- A. diameter
 - B. circumference
 - C. radius
 - D. chord
- 15.** A straight line passing through the centre of a circle and joining two points on the circle is the
- A. radius
 - B. tangent
 - C. arc
 - D. diameter
- 16.** If r is the radius and d the diameter of a circle, then
- A. $d = r$
 - B. $d = 2r$
 - C. $d = r \div 2$
 - D. $d = r + 2$
- 17.** The ratio of the circumference of a circle to its diameter is always
- A. π (pi)
 - B. 2
 - C. 4
 - D. the radius
- 18.** A good fractional approximation of π used in schools is
- A. $22/5$
 - B. $21/7$
 - C. $22/7$
 - D. $7/22$
- 19.** The circumference C of a circle of radius r is given by
- A. $C = \pi r^2$
 - B. $C = \pi r$
 - C. $C = 2\pi + r$
 - D. $C = 2\pi r$
- 20.** Find the circumference of a circle of radius 7 cm. [Take $\pi = 22/7$]
- A. 22 cm
 - B. 44 cm
 - C. 154 cm
 - D. 88 cm

- 21.** Find the circumference of a circle of diameter 10 cm, rounding your answer to the nearest tenth. [Take $\pi = 3.142$]
- A. 31.4 cm
 - B. 62.8 cm
 - C. 314.2 cm
 - D. 3.14 cm
- 22.** The circumference of a circle is 88 cm. Find its diameter. [Take $\pi = 22/7$]
- A. 14 cm
 - B. 44 cm
 - C. 28 cm
 - D. 176 cm
- 23.** The diameter of a circle is 140 cm. Its radius is
- A. 280 cm
 - B. 70 cm
 - C. 140 cm
 - D. 35 cm
- 24.** The radius of a circle is 140 cm. Find its circumference. [Take $\pi = 22/7$]
- A. 440 cm
 - B. 1760 cm
 - C. 220 cm
 - D. 880 cm
- 25.** The radius of a circle is 140 cm. Its diameter is
- A. 280 cm
 - B. 70 cm
 - C. 440 cm
 - D. 140 cm
- 26.** A tin lid has circumference 38 cm and diameter 12 cm. The value of $c \div d$, correct to 2 decimal places, is
- A. 0.32
 - B. 26
 - C. 3.17
 - D. 3.8
- 27.** Which of the following can be used to measure the circumference of a can directly?
- A. a balance
 - B. a protractor
 - C. a straight ruler alone
 - D. a piece of string and a ruler

- 28.** A part of the circumference of a circle between two points on it is called
- A. an arc
 - B. a sector
 - C. a diameter
 - D. a centre
- 29.** The region of a circle enclosed by two radii and an arc is called
- A. a chord
 - B. a sector
 - C. a segment
 - D. a tangent
- 30.** A circular roundabout has radius 14 m. A man walks once round it. The distance he walks is [Take $\pi = 22/7$]
- A. 44 m
 - B. 28 m
 - C. 88 m
 - D. 616 m
- 31.** A bicycle wheel has radius 35 cm. The distance covered in one complete turn of the wheel is [Take $\pi = 22/7$]
- A. 110 cm
 - B. 70 cm
 - C. 440 cm
 - D. 220 cm
- 32.** The circumference of a car wheel is 132 cm. Find the radius of the wheel. [Take $\pi = 22/7$]
- A. 42 cm
 - B. 21 cm
 - C. 84 cm
 - D. 66 cm
- 33.** On a square grid, the area of a rectangle is found by
- A. length + width
 - B. $2 \times (\text{length} + \text{width})$
 - C. $4 \times \text{length}$
 - D. length $\times$ width
- 34.** A rectangle drawn on a square grid is 4 units long and 3 units wide. Its area is
- A. 12 square units
 - B. 7 square units
 - C. 14 square units
 - D. 24 square units

- 35.** A rectangle drawn on a square grid is 4 units by 3 units. Another rectangle is drawn with twice the area. The area of the new rectangle is
- A. 6 square units
 - B. 12 square units
 - C. 24 square units
 - D. 48 square units
- 36.** A rectangle on a square grid is 4 units wide and 6 units long. A new rectangle is twice as wide and one and a half times as long. Its dimensions are
- A. 6 units wide and 9 units long
 - B. 8 units wide and 9 units long
 - C. 8 units wide and 6 units long
 - D. 4 units wide and 9 units long
- 37.** The area of a triangle is given by
- A. $\frac{1}{2} \times \text{base} \times \text{height}$
 - B. $\text{base} \times \text{height}$
 - C. $2 \times \text{base} \times \text{height}$
 - D. $\text{base} + \text{height}$
- 38.** A triangle has base 6 units and height 4 units. Its area is
- A. 24 square units
 - B. 10 square units
 - C. 12 square units
 - D. 6 square units
- 39.** Which base-and-height pair can be used to draw a triangle of area 3 square units on a square grid?
- A. base 3 units, height 3 units
 - B. base 2 units, height 2 units
 - C. base 6 units, height 2 units
 - D. base 6 units, height 1 unit
- 40.** How many different triangles of the same area can be drawn on a square grid when the base and height stay the same?
- A. many different triangles
 - B. only one triangle
 - C. exactly two triangles
 - D. no triangle

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) Perimeters in mixed units.

Question 2

- (a) Parts of a circle and the value of π .

Question 3

- (a) Circumference of a circle.

Question 4

- (a) Solving problems with circumference.

Question 5

- (a) Drawing on square grids.

Question 6

- (a) Mixed problems.

Chapter 3: Ch12

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. A rectangle is cut along a diagonal to form two triangles. The area of each triangle is
 - A. half the area of the rectangle
 - B. equal to the area of the rectangle
 - C. twice the area of the rectangle
 - D. one quarter of the area of the rectangle
2. The area of a triangle is given by
 - A. $\text{base} \times \text{height}$
 - B. $2 \times \text{base} \times \text{height}$
 - C. $\text{base} + \text{height}$
 - D. $\frac{1}{2} \times \text{base} \times \text{height}$
3. A triangle has base 8 cm and perpendicular height 5 cm. The area of the rectangle that encloses it is
 - A. 20 cm^2
 - B. 40 cm^2
 - C. 13 cm^2
 - D. 80 cm^2
4. The area of the triangle in Question 3 is
 - A. 40 cm^2
 - B. 13 cm^2
 - C. 20 cm^2
 - D. 80 cm^2
5. In the formula $\frac{1}{2} \times \text{base} \times \text{height}$, the height used is the
 - A. perpendicular height
 - B. length of a slant side
 - C. perimeter
 - D. longest side
6. A triangle and a parallelogram have the same base and the same height. The area of the triangle is
 - A. equal to that of the parallelogram
 - B. twice that of the parallelogram
 - C. half that of the parallelogram

D. one quarter of that of the parallelogram

- 7.** The formula for the area of a triangle includes $\frac{1}{2}$ because
- A. triangles are always small
 - B. a triangle is half of a rectangle or parallelogram with the same base and height
 - C. areas are always halved
 - D. the height is measured twice
- 8.** The area of a parallelogram with base b and height h is
- A. $\frac{1}{2} b h$
 - B. $b + h$
 - C. $2 b h$
 - D. $b h$
- 9.** Two identical triangles can be fitted together to form
- A. a parallelogram
 - B. a circle
 - C. a pentagon
 - D. a cube
- 10.** A triangle drawn on square grid encloses 6 whole unit squares and 8 half unit squares. Its area is
- A. 14 square units
 - B. 12 square units
 - C. 10 square units
 - D. 8 square units
- 11.** Which of the following is NOT needed to find the area of a triangle?
- A. its base
 - B. its perpendicular height
 - C. the factor $\frac{1}{2}$
 - D. the length of its slant side
- 12.** The area of a right-angled triangle whose two perpendicular sides are 6 cm and 8 cm is
- A. 24 cm^2
 - B. 48 cm^2
 - C. 14 cm^2
 - D. 28 cm^2

- 13.** Find the area of a triangle with base 10 cm and height 6 cm.
- A. 60 cm^2
 - B. 16 cm^2
 - C. 30 cm^2
 - D. 32 cm^2
- 14.** Find the area of a triangle with base 12 cm and height 9 cm.
- A. 108 cm^2
 - B. 54 cm^2
 - C. 21 cm^2
 - D. 42 cm^2
- 15.** Find the area of a triangle with base 7 cm and height 4 cm.
- A. 28 cm^2
 - B. 11 cm^2
 - C. 22 cm^2
 - D. 14 cm^2
- 16.** A triangular garden has base 15 m and height 8 m. Its area is
- A. 120 m^2
 - B. 60 m^2
 - C. 23 m^2
 - D. 46 m^2
- 17.** The area of a triangle is 24 cm^2 and its base is 8 cm. Its perpendicular height is
- A. 3 cm
 - B. 12 cm
 - C. 6 cm
 - D. 4 cm
- 18.** The area of a triangle is 50 cm^2 and its perpendicular height is 10 cm. Its base is
- A. 10 cm
 - B. 5 cm
 - C. 25 cm
 - D. 100 cm
- 19.** Find the area of a triangle with base 4.2 cm and height 5 cm.
- A. 21 cm^2
 - B. 9.2 cm^2
 - C. 42 cm^2
 - D. 10.5 cm^2

- 20.** Find the area of a triangle with base 9 cm and height 6 cm.
- A. 54 cm^2
 - B. 27 cm^2
 - C. 15 cm^2
 - D. 30 cm^2
- 21.** A triangle has area 36 cm^2 and base 9 cm. Its perpendicular height is
- A. 8 cm
 - B. 4 cm
 - C. 16 cm
 - D. 6 cm
- 22.** Find the area of a triangle with base 3.5 cm and height 4 cm.
- A. 14 cm^2
 - B. 7.5 cm^2
 - C. 3.5 cm^2
 - D. 7 cm^2
- 23.** The area of a right-angled triangle whose two perpendicular sides are 5 cm and 12 cm is
- A. 60 cm^2
 - B. 30 cm^2
 - C. 17 cm^2
 - D. 34 cm^2
- 24.** A triangular flower bed has base 20 m and height 15 m. Its area is
- A. 150 m^2
 - B. 300 m^2
 - C. 35 m^2
 - D. 70 m^2
- 25.** A triangle has area 42 cm^2 and perpendicular height 7 cm. Its base is
- A. 6 cm
 - B. 21 cm
 - C. 12 cm
 - D. 24 cm
- 26.** A triangle has area 10 cm^2 . If both its base and its height are doubled, the new area is
- A. 20 cm^2
 - B. 10 cm^2
 - C. 15 cm^2
 - D. 40 cm^2

- 27.** A triangular flag has base 30 cm and height 20 cm. The area of cloth needed for the flag is
- A. 600 cm^2
 - B. 300 cm^2
 - C. 50 cm^2
 - D. 100 cm^2
- 28.** A triangular plot of land has base 40 m and height 25 m. Its area is
- A. 1000 m^2
 - B. 65 m^2
 - C. 500 m^2
 - D. 130 m^2
- 29.** A triangular piece of metal has area 60 cm^2 and base 12 cm. Its perpendicular height is
- A. 5 cm
 - B. 24 cm
 - C. 6 cm
 - D. 10 cm
- 30.** A triangular wall has base 10 m and height 6 m. Paint costs GH¢2.00 per square metre. The cost of painting the wall is
- A. GH¢60.00
 - B. GH¢30.00
 - C. GH¢120.00
 - D. GH¢16.00
- 31.** A square of side 8 cm is cut along a diagonal. The area of each triangle formed is
- A. 64 cm^2
 - B. 32 cm^2
 - C. 16 cm^2
 - D. 8 cm^2
- 32.** A rectangle 12 cm by 5 cm is cut along a diagonal. The area of each triangle is
- A. 30 cm^2
 - B. 60 cm^2
 - C. 17 cm^2
 - D. 25 cm^2
- 33.** Two triangles each have base 6 cm and height 4 cm. Their total area is
- A. 12 cm^2
 - B. 48 cm^2
 - C. 20 cm^2
 - D. 24 cm^2

- 34.** The triangular sail of a boat has base 4 m and height 3 m. The area of the sail is
- A. 12 m^2
 - B. 7 m^2
 - C. 6 m^2
 - D. 24 m^2
- 35.** A shape is made of a rectangle 10 cm by 4 cm with a triangle of base 10 cm and height 3 cm on top. The total area is
- A. 40 cm^2
 - B. 55 cm^2
 - C. 70 cm^2
 - D. 47.5 cm^2
- 36.** A triangle has area 27 cm^2 and perpendicular height 6 cm. Its base is
- A. 9 cm
 - B. 4.5 cm
 - C. 18 cm
 - D. 3 cm
- 37.** A farmer has a triangular field with base 60 m and height 50 m. The area of the field is
- A. 3000 m^2
 - B. 110 m^2
 - C. 1500 m^2
 - D. 220 m^2
- 38.** Four identical triangular flags, each of base 20 cm and height 15 cm, are to be made. The total area of cloth needed is
- A. 150 cm^2
 - B. 300 cm^2
 - C. 1200 cm^2
 - D. 600 cm^2
- 39.** Two triangles have the same base and the same height. They must
- A. be congruent
 - B. have equal areas
 - C. have equal perimeters
 - D. be right-angled
- 40.** The area of a triangle with base 2.4 m and height 5 m is
- A. 6 m^2
 - B. 12 m^2
 - C. 7.4 m^2
 - D. 3 m^2

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) Deriving the formula for the area of a triangle.

Question 2

- (a) Calculating areas of triangles.

Question 3

- (a) Finding missing dimensions.

Question 4

- (a) Areas on square grids.

Question 5

- (a) Area problems in real life.

Question 6

- (a) Creating and reasoning.

Chapter 4: Ch1

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Bearings are always measured from the direction
 - A. East
 - B. South
 - C. West
 - D. North
2. Bearings are measured in the
 - A. clockwise direction
 - B. anti-clockwise direction
 - C. either direction
 - D. vertical direction
3. A bearing is written as a
 - A. one-digit number
 - B. two-digit number
 - C. three-digit number
 - D. four-digit number
4. The bearing of a point due East of another point is
 - A. 045°
 - B. 090°
 - C. 180°
 - D. 270°
5. The bearing of a point due South of another point is
 - A. 090°
 - B. 270°
 - C. 300°
 - D. 180°
6. The bearing of a point due West of another point is
 - A. 270°
 - B. 180°
 - C. 090°
 - D. 360°

- 7.** The bearing of a point due North of another point is written as
- A. 090°
 - B. 000°
 - C. 180°
 - D. 045°
- 8.** From a point O, a point P is 60° measured clockwise from the North. The three-figure bearing of P from O is
- A. 030°
 - B. 300°
 - C. 120°
 - D. 060°
- 9.** The bearing of P from Q is 060° . The bearing of Q from P is
- A. 120°
 - B. 060°
 - C. 240°
 - D. 300°
- 10.** The bearing of P from Q is 145° . The bearing of Q from P is
- A. 325°
 - B. 035°
 - C. 215°
 - D. 055°
- 11.** The bearing of A from B is 200° . The bearing of B from A is
- A. 380°
 - B. 100°
 - C. 020°
 - D. 110°
- 12.** The back bearing of a bearing less than 180° is found by
- A. subtracting it from 360°
 - B. multiplying it by 2
 - C. subtracting 90°
 - D. adding 180°
- 13.** The back bearing of a bearing greater than 180° is found by
- A. subtracting 180°
 - B. adding 180°
 - C. subtracting it from 90°
 - D. adding 90°

- 14.** The bearing of X from Y is 275° . The bearing of Y from X is
- A. 455°
 - B. 095°
 - C. 085°
 - D. 105°
- 15.** The bearing of B from A is 040° and $AB = 5$ km. This information describes
- A. a scalar quantity only
 - B. the speed of travel
 - C. the position of B relative to A
 - D. the area between A and B
- 16.** Two towns P and Q are 8 km apart. The bearing of Q from P is 120° . The bearing of P from Q is
- A. 060°
 - B. 300°
 - C. 240°
 - D. 120°
- 17.** A vector quantity has
- A. magnitude and direction
 - B. magnitude only
 - C. direction only
 - D. neither magnitude nor direction
- 18.** Which of the following is a vector quantity?
- A. speed
 - B. distance
 - C. velocity
 - D. mass
- 19.** Which of the following is a scalar quantity?
- A. force
 - B. weight
 - C. velocity
 - D. time
- 20.** Which list contains scalar quantities only?
- A. force, mass, time
 - B. distance, speed, mass
 - C. velocity, energy, work
 - D. weight, momentum, volume

- 21.** Which list contains vector quantities only?
- A. weight, force, velocity, momentum
 - B. time, force, speed, mass
 - C. distance, energy, velocity, work
 - D. mass, volume, weight, energy
- 22.** A vector may be identified as a movement (distance) along
- A. any curved path
 - B. the North line only
 - C. a given bearing
 - D. a circle
- 23.** The vector (3 km, 060°) means
- A. 3 km at a bearing of 030°
 - B. a movement of 3 km on a bearing of 060°
 - C. 60 km on a bearing of 003°
 - D. 3 km due North
- 24.** In the vector (5 km, 120°), the direction of movement is on a bearing of
- A. 060°
 - B. 030°
 - C. 210°
 - D. 120°
- 25.** The column form of a vector with x-component 2 and y-component 3 is
- A. (2, 3)
 - B. (3, 2)
 - C. (5, 0)
 - D. (2, -3)
- 26.** In the column vector (x, y), the x-component represents movement in the
- A. vertical direction
 - B. direction of the bearing
 - C. magnitude of the vector
 - D. horizontal (East-West) direction
- 27.** In the column vector (x, y), the y-component represents movement in the
- A. horizontal direction
 - B. vertical (North-South) direction
 - C. speed of travel
 - D. total distance

- 28.** The column vector $(-3, 4)$ represents a movement of
- A. 3 units East and 4 units North
 - B. 3 units West and 4 units South
 - C. 3 units West and 4 units North
 - D. 4 units West and 3 units North
- 29.** The magnitude of the vector $(3, 4)$ is
- A. 7
 - B. 12
 - C. 25
 - D. 5
- 30.** The magnitude of the vector $(6, 8)$ is
- A. 10
 - B. 14
 - C. 48
 - D. 100
- 31.** The magnitude of the vector $(5, 12)$ is
- A. 17
 - B. 13
 - C. 60
 - D. 169
- 32.** The magnitude of the vector $(-3, 4)$ is
- A. 1
 - B. -5
 - C. 5
 - D. 7
- 33.** The magnitude of a column vector is found using
- A. a protractor
 - B. the compass rose
 - C. simple addition
 - D. Pythagoras' theorem
- 34.** The magnitude of the vector $(9, 12)$ is
- A. 15
 - B. 21
 - C. 108
 - D. 225

- 35.** The magnitude of the vector (8, 15) is
- A. 23
 - B. 17
 - C. 120
 - D. 7
- 36.** The magnitude of the vector (2, 3), correct to 1 decimal place, is
- A. 3.6
 - B. 5
 - C. 13
 - D. 2.5
- 37.** A boat sails 6 km North and then 8 km East. Its straight-line distance from the starting point is
- A. 14 km
 - B. 48 km
 - C. 2 km
 - D. 10 km
- 38.** A man walks 9 km East and then 12 km South. His straight-line distance from the start is
- A. 21 km
 - B. 15 km
 - C. 108 km
 - D. 3 km
- 39.** A girl walks 4 km North and then 3 km East. Her straight-line distance from the start is
- A. 7 km
 - B. 12 km
 - C. 5 km
 - D. 1 km
- 40.** The magnitude of the vector (7, 24) is
- A. 25
 - B. 31
 - C. 168
 - D. 625

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Bearings.

Question 2

(a) Back bearings.

Question 3

(a) Scalar and vector quantities.

Question 4

(a) Vectors in column form.

Question 5

(a) Magnitude of vectors.

Question 6

(a) Mixed problems.

Chapter 5: Ch14

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. A shape has reflectional (fold) symmetry if it can be
 - A. moved to a new position
 - B. folded into two matching halves
 - C. turned through any angle
 - D. made bigger
2. The number of lines of symmetry of a square is
 - A. 1
 - B. 2
 - C. 4
 - D. 8
3. The number of lines of symmetry of a rectangle (which is not a square) is
 - A. 2
 - B. 4
 - C. 1
 - D. 0
4. Which of the following has NO line of symmetry?
 - A. an equilateral triangle
 - B. an isosceles triangle
 - C. a circle
 - D. a scalene triangle
5. Many adinkra symbols are examples of designs with
 - A. translation only
 - B. reflectional symmetry
 - C. no symmetry at all
 - D. curved edges only
6. The number of lines of symmetry of a circle is
 - A. 1
 - B. 4
 - C. 0
 - D. unlimited

- 7.** The number of lines of symmetry of an equilateral triangle is
- A. 3
 - B. 0
 - C. 1
 - D. 2
- 8.** The fold line that divides a symmetric shape into two matching halves is called a
- A. diagonal
 - B. line of symmetry
 - C. radius
 - D. tangent
- 9.** The reflection of the point (3, 1) in the x-axis is
- A. (-3, 1)
 - B. (1, 3)
 - C. (3, -1)
 - D. (-3, -1)
- 10.** The reflection of the point (2, 5) in the y-axis is
- A. (2, -5)
 - B. (5, 2)
 - C. (-2, -5)
 - D. (-2, 5)
- 11.** Under reflection, a shape and its image are always
- A. congruent
 - B. of different sizes
 - C. of different shapes
 - D. rotated
- 12.** The reflection of the point (4, 3) in the x-axis is
- A. (3, 4)
 - B. (-4, 3)
 - C. (4, -3)
 - D. (-4, -3)
- 13.** The reflection of the point (-2, 3) in the y-axis is
- A. (-2, -3)
 - B. (3, -2)
 - C. (-2, 3)
 - D. (2, 3)

- 14.** A point that lies on the mirror line, when reflected,
- A. moves to the opposite side
 - B. stays in the same position
 - C. moves twice as far
 - D. becomes the origin
- 15.** A point and its reflection in a mirror line are
- A. equal distances from the mirror line on opposite sides
 - B. on the same side of the mirror line
 - C. at different distances from the mirror line
 - D. both on the mirror line
- 16.** A point is 3 units to the right of the origin and 2 units up. Its coordinates are
- A. (2, 3)
 - B. (-3, 2)
 - C. (3, 2)
 - D. (3, -2)
- 17.** The point A(3, 1) is reflected in the x-axis. The image of A is
- A. (1, 3)
 - B. (3, -1)
 - C. (-3, 1)
 - D. (-3, -1)
- 18.** The triangle with vertices (1, 1), (3, 1) and (1, 4) is reflected in the y-axis. The vertices of the image are
- A. (-1, 1), (-3, 1), (-1, 4)
 - B. (1, -1), (3, -1), (1, -4)
 - C. (-1, -1), (-3, -1), (-1, -4)
 - D. (1, 1), (3, 1), (1, 4)
- 19.** The reflection of the point (0, 5) in the x-axis is
- A. (5, 0)
 - B. (0, 5)
 - C. (5, 5)
 - D. (0, -5)
- 20.** Reflection in the x-axis changes
- A. both coordinates
 - B. neither coordinate
 - C. the sign of the y-coordinate
 - D. the sign of the x-coordinate

- 21.** A translation is
- A. flipping a shape over a line
 - B. turning a shape about a point
 - C. changing the size of a shape
 - D. sliding a shape without turning or flipping it
- 22.** People going down a slide is a real-life example of
- A. translation
 - B. reflection
 - C. enlargement
 - D. symmetry
- 23.** The point $(2, 3)$ is translated by the vector $(4, 1)$. Its image is
- A. $(6, 3)$
 - B. $(8, 4)$
 - C. $(6, 4)$
 - D. $(-2, 2)$
- 24.** The point $(5, 2)$ is translated by the vector $(-3, 2)$. Its image is
- A. $(8, 4)$
 - B. $(2, 4)$
 - C. $(2, 0)$
 - D. $(-3, 2)$
- 25.** The point $(1, 1)$ is translated by the vector $(2, 5)$. Its image is
- A. $(3, 5)$
 - B. $(2, 5)$
 - C. $(5, 2)$
 - D. $(3, 6)$
- 26.** The point $(4, -2)$ is translated by the vector $(-1, 3)$. Its image is
- A. $(3, 1)$
 - B. $(5, 1)$
 - C. $(3, -5)$
 - D. $(-1, 3)$
- 27.** A shape translated by the vector $(0, 4)$ moves
- A. 4 units left
 - B. 4 units up
 - C. 4 units right
 - D. 4 units down

- 28.** After a translation, the image is
- A. a different shape from the object
 - B. smaller than the object
 - C. bigger than the object
 - D. congruent to the object
- 29.** The translation that moves the point (2, 2) to (5, 6) is the vector
- A. (3, 4)
 - B. (7, 8)
 - C. (-3, -4)
 - D. (4, 3)
- 30.** The translation that moves the point (3, 5) to (1, 2) is the vector
- A. (2, 3)
 - B. (4, 7)
 - C. (-2, -3)
 - D. (-3, -2)
- 31.** Congruent shapes are shapes which have
- A. the same shape only
 - B. the same size only
 - C. different shapes
 - D. the same shape and the same size
- 32.** Similar shapes are shapes which have
- A. the same size
 - B. different shapes
 - C. the same shape but different sizes
 - D. the same shape and the same size
- 33.** Which of the following statements is true?
- A. All congruent shapes are similar
 - B. All similar shapes are congruent
 - C. Similar shapes must be congruent
 - D. Congruent shapes have different sizes
- 34.** Carpet designs and fabric patterns often contain
- A. only single shapes
 - B. repeated congruent shapes
 - C. no shapes at all
 - D. circles only

- 35.** A rectangle 4 cm by 2 cm and a rectangle 8 cm by 4 cm are
- A. congruent
 - B. neither similar nor congruent
 - C. congruent but not similar
 - D. similar but not congruent
- 36.** Two circles of different radii are always
- A. similar
 - B. congruent
 - C. neither similar nor congruent
 - D. equal in area
- 37.** A photograph is enlarged by a photocopier. The enlargement and the original photograph are
- A. congruent
 - B. neither similar nor congruent
 - C. different in shape
 - D. similar
- 38.** Two triangles are congruent. One has sides 3 cm, 4 cm and 5 cm. The sides of the other are
- A. 6 cm, 8 cm, 10 cm
 - B. 3 cm, 4 cm, 5 cm
 - C. 4 cm, 5 cm, 6 cm
 - D. 3 cm, 3 cm, 3 cm
- 39.** A triangle with sides 2 cm, 3 cm and 4 cm is similar to a triangle with sides
- A. 2 cm, 3 cm, 5 cm
 - B. 3 cm, 4 cm, 5 cm
 - C. 4 cm, 6 cm, 8 cm
 - D. 2 cm, 4 cm, 6 cm
- 40.** Under reflection and under translation, the object and its image are always
- A. congruent
 - B. similar but not congruent
 - C. of different sizes
 - D. turned

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Reflectional (fold) symmetry.

Question 2

(a) Reflection in the coordinate plane.

Question 3

(a) Translation by vectors.

Question 4

(a) Congruent and similar shapes.

Question 5

(a) Mixed coordinate transformations.

Question 6

(a) Creating and reasoning.

Chapter 6: Ch1

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Data that can be counted or measured and expressed as numbers is called
 - A. qualitative data
 - B. quantitative data
 - C. grouped data
 - D. secondary data
2. The favourite subject of a JHS 1 student is an example of
 - A. quantitative data
 - B. continuous data
 - C. qualitative data
 - D. discrete data
3. A class wants to find the favourite drink of its students so it can buy drinks for a party. The best method of collecting this data is a
 - A. observation
 - B. experiment
 - C. database search
 - D. questionnaire
4. The best method of finding the number of desks in each classroom of a school is
 - A. observation and counting
 - B. interview
 - C. questionnaire
 - D. internet search
5. Collecting data by watching people or events and recording what happens, without asking questions, is called
 - A. experiment
 - B. observation
 - C. database search
 - D. survey

- 6.** Which method of data collection makes use of already recorded information such as registers, books or the internet?
- A. database / electronic media
 - B. observation
 - C. interview
 - D. experiment
- 7.** Which of the following is quantitative data?
- A. the favourite hobby of a student
 - B. the favourite day of the week
 - C. opinions about school meals
 - D. the height of students in a class
- 8.** To find out whether people who eat more fufu develop pot belly, the best method of collecting data is a
- A. survey
 - B. questionnaire
 - C. experiment
 - D. database search
- 9.** Data collected directly from the people concerned by asking them questions is called
- A. secondary data
 - B. grouped data
 - C. qualitative data
 - D. primary data
- 10.** Data obtained from sources such as textbooks, newspapers or the internet, rather than collected firsthand, is called
- A. primary data
 - B. secondary data
 - C. raw data
 - D. discrete data
- 11.** The best method of finding the amount of money Basic 6 pupils spend on bus fare to school every month is a
- A. questionnaire
 - B. observation
 - C. experiment
 - D. simulation
- 12.** Which of the following questions will collect quantitative data?
- A. What is your favourite colour?
 - B. What is your favourite subject?
 - C. How many brothers and sisters do you have?
 - D. Do you enjoy homework?

- 13.** A questionnaire is
- A. a type of graph
 - B. a set of written questions used to collect data from people
 - C. a table of frequencies
 - D. a picture used to represent data
- 14.** Which of the following is a good question for a questionnaire about favourite subjects?
- A. Tell us your whole school history
 - B. Don't you like mathematics?
 - C. What is your favourite subject?
 - D. Is school food nice?
- 15.** A good questionnaire question should be
- A. clear and easy to understand
 - B. leading, pushing the person to one answer
 - C. long and confusing
 - D. two questions joined together
- 16.** The question "Don't you think mathematics is the best subject?" is not a good questionnaire question because it
- A. is too short
 - B. pushes the respondent towards one answer
 - C. cannot be answered at all
 - D. collects only numbers
- 17.** Which of the following is the best way to ask about a pupil's favourite snack in a questionnaire?
- A. Write an essay about food
 - B. Describe every meal you have ever eaten
 - C. Tick one: mango, biscuit, groundnut, other
 - D. What about food?
- 18.** To administer a questionnaire means to
- A. draw a graph of the results
 - B. give the questionnaire to respondents to answer
 - C. count the tally marks
 - D. calculate the sector angles
- 19.** After the responses to a questionnaire have been collected, the next step is to
- A. organise and record the results in a table
 - B. throw away the responses
 - C. design another questionnaire
 - D. draw the pie chart first

- 20.** In a class survey, the question “What is your favourite day of the week?” collects
- A. quantitative data
 - B. continuous data
 - C. qualitative data
 - D. grouped data
- 21.** The question “How much do you spend on bus fare to school every day?” collects
- A. qualitative data
 - B. quantitative data
 - C. secondary data
 - D. opinions
- 22.** The best way to record the results collected with a questionnaire is in a
- A. frequency (tally) table
 - B. long essay
 - C. list of names only
 - D. photograph
- 23.** Which of the following is NOT a feature of a good questionnaire?
- A. clear questions
 - B. space for the respondent’s answers
 - C. leading or ambiguous questions
 - D. short and focused questions
- 24.** In a survey of 30 students, 12 said Mathematics is their favourite subject. What percentage of the students chose Mathematics?
- A. 12%
 - B. 30%
 - C. 25%
 - D. 40%
- 25.** When counting with tally marks, the fifth mark is drawn
- A. vertically, next to the others
 - B. horizontally below the group
 - C. diagonally across the four marks
 - D. in a new group far away
- 26.** The following data were recorded: 4, 2, 4, 3, 2, 4, 1, 3, 4, 2. The frequency of the value 4 is
- A. 2
 - B. 3
 - C. 5
 - D. 4

- 27.** The total number of values in the data set 4, 2, 4, 3, 2, 4, 1, 3, 4, 2 is
- A. 10
 - B. 8
 - C. 12
 - D. 9
- 28.** In a frequency table, the frequencies of the marks 1, 2, 3, 4 and 5 are 2, 6, 4, 5 and 3 respectively. How many students took the test?
- A. 15
 - B. 20
 - C. 25
 - D. 18
- 29.** Using the frequency table in question 28, the mark obtained by the greatest number of students is
- A. 3
 - B. 5
 - C. 2
 - D. 4
- 30.** Using the frequency table in question 28, the number of students who scored 3 or more is
- A. 12
 - B. 10
 - C. 14
 - D. 15
- 31.** The best way to display the change in temperature over a week is a
- A. pie chart
 - B. pictograph
 - C. line graph
 - D. bar chart
- 32.** The best way to compare the quantities of different categories of data is a
- A. line graph
 - B. bar chart
 - C. pie chart
 - D. tally table
- 33.** Which display uses sectors of a circle to represent parts of the data?
- A. bar chart
 - B. line graph
 - C. pictograph
 - D. pie chart

- 34.** Which display uses pictures or symbols to represent numbers?
- A. pictograph
 - B. pie chart
 - C. bar chart
 - D. line graph
- 35.** In a pictograph, one symbol represents 5 students. How many symbols are needed to represent 35 students?
- A. 5
 - B. 7
 - C. 6
 - D. 8
- 36.** In a survey of 40 students, 16 chose football as their favourite sport. The sector angle for football in a pie chart is
- A. 120°
 - B. 160°
 - C. 144°
 - D. 90°
- 37.** Using the survey in question 36, 10 students chose volleyball. The sector angle for volleyball is
- A. 100°
 - B. 72°
 - C. 54°
 - D. 90°
- 38.** In a bar chart, the height of each bar represents the
- A. frequency
 - B. mean
 - C. sector angle
 - D. range
- 39.** A double bar chart is used for
- A. counting frequencies
 - B. showing sector angles
 - C. drawing symbols
 - D. comparing two sets of data
- 40.** For a pictograph to be read correctly, it must include
- A. a key stating what each symbol represents
 - B. sector angles
 - C. equal bars
 - D. a circle

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Data collection.

Question 2

(a) Questionnaires.

Question 3

(a) From questionnaire to graph.

Question 4

(a) Reading a frequency table.

Question 5

(a) Pie chart.

Question 6

(a) Creating and reasoning.

Chapter 7: Ch16

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. The mean of a data set is found by
 - A. taking the value that occurs most often
 - B. taking the middle value when the data is arranged in order
 - C. dividing the sum of all the values by the number of values
 - D. subtracting the smallest value from the largest
2. The mean of the data set {8, 9, 7, 6, 8, 10} is
 - A. 8
 - B. 6
 - C. 7
 - D. 9
3. The sum of five numbers is 60. The mean of the numbers is
 - A. 55
 - B. 300
 - C. 5
 - D. 12
4. The mean of 4, 8 and 6 is
 - A. 4
 - B. 6
 - C. 8
 - D. 18
5. The mean of five numbers is 10. The sum of the five numbers is
 - A. 50
 - B. 15
 - C. 2
 - D. 10
6. Ama scored 7, 8, 6 and 9 in four tests. Her mean score is
 - A. 7
 - B. 7.5
 - C. 8
 - D. 30

- 7.** The mean of 2, 4, 6, 8 and 10 is
- A. 5
 - B. 30
 - C. 6
 - D. 8
- 8.** If every value in a data set is increased by 3, the mean of the data set
- A. increases by 3
 - B. remains the same
 - C. increases by 6
 - D. becomes 3
- 9.** The marks obtained by 20 students in a test are shown below: Mark 1 → 2 students, Mark 2 → 6 students, Mark 3 → 4 students, Mark 4 → 5 students, Mark 5 → 3 students. The mean mark is
- A. 3
 - B. 2.8
 - C. 3.5
 - D. 3.05
- 10.** The mean age of a group of children is 12 years and the sum of their ages is 96 years. How many children are in the group?
- A. 12
 - B. 8
 - C. 96
 - D. 108
- 11.** A baker sold 20, 25, 30, 24 and 26 loaves of bread on five successive days. The mean number of loaves sold per day is
- A. 24
 - B. 26
 - C. 30
 - D. 25
- 12.** The mean of four numbers is 15. Three of the numbers are 12, 16 and 14. The fourth number is
- A. 12
 - B. 15
 - C. 18
 - D. 60

- 13.** Which measure of central tendency is most affected by a very large value (outlier) in the data?
- A. mean
 - B. median
 - C. mode
 - D. none of them
- 14.** Which measure of central tendency uses every value in the data set when it is being calculated?
- A. mode
 - B. mean
 - C. median
 - D. range
- 15.** Kofi's bus fares to school on four days were GH¢3, GH¢2, GH¢4 and GH¢3. The mean daily fare is
- A. GH¢2
 - B. GH¢2.50
 - C. GH¢4
 - D. GH¢3
- 16.** Sena's scores in five subjects are 75, 87, 90, 88 and 79. If she wants a mean score of 85 over six subjects, what must she score in the sixth subject?
- A. 91
 - B. 85
 - C. 90
 - D. 89
- 17.** The median of a data set is
- A. the value that occurs most often
 - B. the middle value when the values are arranged in order
 - C. the sum of the values divided by 2
 - D. the largest value in the data set
- 18.** The first step in finding the median of a data set is to
- A. add all the values
 - B. find the mode
 - C. draw a bar chart
 - D. arrange the values in ascending or descending order
- 19.** The median of 19, 29, 36, 15 and 20 is
- A. 15
 - B. 36
 - C. 20
 - D. 19

- 20.** The median of the data set 8, 9, 7, 6, 8, 10 is
- A. 8
 - B. 8.5
 - C. 7
 - D. 9
- 21.** The median of 5, 2, 9, 1 and 7 is
- A. 2
 - B. 5
 - C. 9
 - D. 7
- 22.** When a data set has an even number of values, the median is found by
- A. taking the larger of the two middle values
 - B. taking the smaller of the two middle values
 - C. taking the mean of the two middle values
 - D. adding all the values
- 23.** The median of 12, 15, 10 and 18 is
- A. 12
 - B. 13.5
 - C. 15
 - D. 14
- 24.** The heights, in cm, of six students are 120, 135, 118, 127, 140 and 130. The median height is
- A. 128.5
 - B. 127
 - C. 130
 - D. 128
- 25.** The table shows the marks obtained by 20 students: Mark 1 → 2 students, Mark 2 → 6 students, Mark 3 → 4 students, Mark 4 → 5 students, Mark 5 → 3 students. The median mark is
- A. 2
 - B. 3.05
 - C. 3
 - D. 4
- 26.** The values 4, 6, 7, 9 and 11 are arranged in order. If the greatest value is changed from 11 to 20, the median
- A. increases to 20
 - B. becomes 8
 - C. decreases
 - D. remains 7

- 27.** For a data set of 20 values arranged in order, the median is the mean of the
- A. 10th and 11th values
 - B. 9th and 10th values
 - C. 10th value only
 - D. 20th value only
- 28.** The median of 2, 2, 2, 2 and 10 is 2. This shows that the median
- A. uses all the values
 - B. is strongly affected by the outlier 10
 - C. is not affected by the extreme value 10
 - D. always equals the mean
- 29.** The mode of a data set is
- A. the middle value
 - B. the value that occurs most often
 - C. the sum of the values
 - D. the average of the values
- 30.** The mode of 2, 3, 3, 5, 3, 6 is
- A. 2
 - B. 5
 - C. 6
 - D. 3
- 31.** A football team scored the following goals in seven matches: 2, 0, 3, 1, 4, 2, 2. The mode of the goals is
- A. 0
 - B. 1
 - C. 2
 - D. 4
- 32.** A data set in which no value occurs more than once has
- A. no mode
 - B. a mode of 0
 - C. every value as its mode
 - D. the mean as its mode
- 33.** The mode(s) of the data set 1, 2, 2, 3, 4, 4, 5 is/are
- A. 2 only
 - B. 4 only
 - C. 3
 - D. 2 and 4

- 34.** The best measure of central tendency to use when finding the most popular drink at a party is the
- A. mean
 - B. mode
 - C. median
 - D. range
- 35.** The table shows the marks obtained by 20 students: Mark 1 $\rightarrow$ 2 students, Mark 2 $\rightarrow$ 6 students, Mark 3 $\rightarrow$ 4 students, Mark 4 $\rightarrow$ 5 students, Mark 5 $\rightarrow$ 3 students. The modal mark is
- A. 2
 - B. 3
 - C. 4
 - D. 5
- 36.** The mean of six numbers is 7. Five of the numbers are 5, 8, 6, 9 and 7. The sixth number is
- A. 6
 - B. 35
 - C. 7
 - D. 8
- 37.** Which measure of central tendency can be used for qualitative data such as favourite colours?
- A. mean
 - B. median
 - C. range
 - D. mode
- 38.** The mode of the data 5, 8, 6, 9, 7, 9 is
- A. 8
 - B. 9
 - C. 7.5
 - D. 6
- 39.** The median of 4, 6, 6 and 8 is
- A. 4
 - B. 8
 - C. 6
 - D. 7

- 40.** For the data set 4, 6, 6, 8, which of the following statements is true?
- A. mean = median = mode
 - B. mean is greater than the mode
 - C. the median is 8
 - D. the mode is 4

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) The mean.

Question 2

(a) The median.

Question 3

(a) The mode.

Question 4

(a) Central tendency from a frequency table.

Question 5

(a) Solving problems with the mean.

Question 6

(a) Creating and reasoning.

Chapter 8: Ch17

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Probability is a measure of
 - A. how fast an event happens
 - B. how likely an event is to happen
 - C. the size of an event
 - D. the number of events that happen
2. "The dog will fly tomorrow." This event is
 - A. certain
 - B. possible
 - C. impossible
 - D. equally likely
3. "Ghana will still be an African country tomorrow." This event is
 - A. certain
 - B. impossible
 - C. doubtful
 - D. possible
4. "Someone in this class will be a teacher in the future." This event is
 - A. impossible
 - B. possible
 - C. certain
 - D. sure
5. An event that will definitely happen is described as
 - A. an event that can never happen
 - B. an event that may or may not happen
 - C. certain
 - D. an event with probability 0
6. An event that can never happen is described as
 - A. impossible
 - B. certain
 - C. an event with probability 1
 - D. possible

- 7.** “The day after Monday will be Tuesday.” This event is
- A. possible
 - B. impossible
 - C. certain
 - D. uncertain
- 8.** “A coin lands heads side up.” This event is
- A. certain
 - B. possible
 - C. impossible
 - D. sure
- 9.** A fair die is thrown once. Obtaining the number 7 is
- A. possible
 - B. certain
 - C. impossible
 - D. equally likely
- 10.** Two fair dice are thrown. Obtaining a total of 13 is
- A. impossible
 - B. possible
 - C. certain
 - D. equally likely
- 11.** Two fair dice are thrown. Obtaining a total of 12 is
- A. impossible
 - B. certain
 - C. possible
 - D. equally likely
- 12.** “It will rain in Winneba in the first week of January.” This event is
- A. impossible
 - B. possible
 - C. certain
 - D. sure
- 13.** The sample space of a probability experiment is
- A. the outcomes that occur most often
 - B. only one outcome of the experiment
 - C. the set of all possible outcomes
 - D. the outcomes that cannot happen

- 14.** The sample space for tossing one fair coin is
- A. {H}
 - B. {H, T}
 - C. {H, T, S}
 - D. {T}
- 15.** How many outcomes are there when a fair die is thrown once?
- A. 6
 - B. 5
 - C. 12
 - D. 2
- 16.** The sample space for throwing one fair die is
- A. {1, 2, 3, 4, 5}
 - B. {2, 4, 6}
 - C. {1, 2, 3, 4, 5, 6}
 - D. {1, 3, 5}
- 17.** Two fair coins are tossed together. The number of possible outcomes is
- A. 2
 - B. 3
 - C. 6
 - D. 4
- 18.** Which of the following is the sample space for tossing two fair coins?
- A. HH, HT, TH, TT
 - B. HH, TT only
 - C. H, T only
 - D. HT, TH only
- 19.** A bag contains 3 red and 2 blue counters. One counter is picked at random. The sample space is
- A. {3, 2}
 - B. {red, blue}
 - C. {red}
 - D. {5}
- 20.** A spinner is divided into 5 equal sectors. How many possible outcomes does it have?
- A. 1
 - B. 10
 - C. 25
 - D. 5

- 21.** A fair die is thrown once. The outcomes that make up the event “a number greater than 4” are
- A. $\{4, 5, 6\}$
 - B. $\{5, 6\}$
 - C. $\{6\}$
 - D. $\{1, 2, 3, 4\}$
- 22.** How many even numbers are in the sample space of a fair die?
- A. 2
 - B. 4
 - C. 3
 - D. 6
- 23.** The probability of an event is calculated as
- A. total outcomes $\div$ favourable outcomes
 - B. favourable outcomes $\div$ total outcomes
 - C. favourable outcomes $+$ total outcomes
 - D. favourable outcomes $\times$ total outcomes
- 24.** A fair die is thrown once. The probability of obtaining a 4 is
- A. $1/6$
 - B. $4/6$
 - C. $1/4$
 - D. 4
- 25.** A fair die is thrown once. The probability of obtaining an even number is
- A. $1/3$
 - B. $2/3$
 - C. $1/6$
 - D. $1/2$
- 26.** A fair die is thrown once. The probability of obtaining a multiple of 3 is
- A. $1/2$
 - B. $1/6$
 - C. $1/3$
 - D. $2/3$
- 27.** A fair die is thrown once. The probability of obtaining a factor of 6 is
- A. $1/2$
 - B. $1/3$
 - C. 1
 - D. $2/3$

- 28.** A fair die is thrown once. The probability of obtaining a factor of 8 is
- A. $1/3$
 - B. $1/2$
 - C. $2/3$
 - D. $1/6$
- 29.** A fair die is thrown once. The probability of obtaining a divisor of 30 is
- A. $5/6$
 - B. $2/3$
 - C. $1/2$
 - D. 1
- 30.** A fair die is thrown once. The probability of obtaining a 3 or greater is
- A. $1/2$
 - B. $1/3$
 - C. $5/6$
 - D. $2/3$
- 31.** A fair die is thrown once. The probability of obtaining a 3 or smaller is
- A. $1/2$
 - B. $2/3$
 - C. $1/3$
 - D. $1/6$
- 32.** The probability of an impossible event is
- A. 1
 - B. 0
 - C. $1/2$
 - D. 100
- 33.** The probability of a certain event is
- A. 0
 - B. $1/2$
 - C. 1
 - D. 10
- 34.** A bag contains 3 red and 2 blue counters. One counter is picked at random. The probability that it is red is
- A. $2/5$
 - B. $1/5$
 - C. $3/2$
 - D. $3/5$

- 35.** Express the probability $\frac{1}{4}$ as a decimal.
- A. 0.25
 - B. 0.14
 - C. 0.4
 - D. 0.75
- 36.** The probability of an event is $\frac{1}{2}$. Expressed as a percentage, it is
- A. 25%
 - B. 12%
 - C. 20%
 - D. 50%
- 37.** A probability of $\frac{1}{2}$ expressed as a ratio is
- A. 1 : 2
 - B. 2 : 1
 - C. 1 : 1
 - D. 2 : 3
- 38.** A fair coin is tossed once. The probability of obtaining heads is
- A. $\frac{1}{4}$
 - B. $\frac{1}{3}$
 - C. $\frac{1}{2}$
 - D. 1
- 39.** A spinner has 4 equal sectors, one of which is red. The probability that the spinner lands on red is
- A. $\frac{1}{2}$
 - B. $\frac{1}{4}$
 - C. $\frac{3}{4}$
 - D. $\frac{1}{5}$
- 40.** A bag contains 2 mangoes, 3 oranges and 5 bananas. One fruit is picked at random. The probability that it is an orange is
- A. $\frac{3}{5}$
 - B. $\frac{1}{2}$
 - C. $\frac{2}{10}$
 - D. $\frac{3}{10}$

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Classifying likelihood.

Question 2

(a) Sample space.

Question 3

(a) Calculating probability with a die.

Question 4

(a) Expressing probabilities in different forms.

Question 5

(a) Probability from a bag.

Question 6

(a) Creating and reasoning.

Chapter 9: Ch

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. What is the place value of the digit 7 in the number 4,275,308,619?
 - A. Millions
 - B. Ten millions
 - C. Hundred millions
 - D. Ten billions
2. Which of the following is six billion, three hundred and twenty million, four hundred and fifty thousand, eight hundred and seven written in numerals?
 - A. 6,302,405,807
 - B. 6,302,450,807
 - C. 6,320,450,807
 - D. 6,320,450,870
3. In the number 5,432,167,890, what is the value of the digit 4?
 - A. 40,000,000
 - B. 400,000,000
 - C. 4,000,000,000
 - D. 4,000,000
4. How many millions make one billion?
 - A. 10
 - B. 100
 - C. 1,000
 - D. 10,000
5. The Government of Ghana budgeted GH¢12,406,000,000 for education in the year 2026/27. How is this amount written in words?
 - A. Twelve billion, forty and six million cedis
 - B. Twelve billion, four hundred and sixty million cedis
 - C. One hundred and twenty-four billion, six million cedis
 - D. Twelve billion, four hundred and six million cedis

- 6.** Which digit occupies the hundred millions place in the number 9,807,654,321?
- A. 9
 - B. 8
 - C. 0
 - D. 7
- 7.** Which of the following is the expanded form of 3,045,000,200?
- A. $3,000,000,000 + 45,000,000 + 200$
 - B. $3,000,000,000 + 4,500,000 + 200$
 - C. $3,000,000,000 + 45,000,000 + 2,000$
 - D. $300,000,000 + 45,000,000 + 200$
- 8.** What is the sum of the value of the digit 6 and the value of the digit 2 in the number 2,615,000,000?
- A. 2,060,000,000
 - B. 260,000,000
 - C. 2,600,000,000
 - D. 8
- 9.** Round 7,823,456,000 to the nearest hundred thousand.
- A. 7,823,500,000
 - B. 7,823,400,000
 - C. 7,820,000,000
 - D. 7,823,460,000
- 10.** Round 2,099,964,000 to the nearest ten thousand.
- A. 2,099,970,000
 - B. 2,100,000,000
 - C. 2,099,960,000
 - D. 2,099,900,000
- 11.** Round 5,600,349,200 to the nearest thousand.
- A. 5,600,350,000
 - B. 5,600,340,000
 - C. 5,600,400,000
 - D. 5,600,349,000
- 12.** Round 3,714,082,547 to the nearest hundred.
- A. 3,714,082,600
 - B. 3,714,082,500
 - C. 3,714,083,000
 - D. 3,714,080,000

- 13.** Round 1,258,437,963 to the nearest ten.
- A. 1,258,437,960
 - B. 1,258,437,970
 - C. 1,258,438,000
 - D. 1,258,437,900
- 14.** A number rounded to the nearest hundred thousand gives 2,400,000,000. Which of the following could be the number?
- A. 2,399,949,000
 - B. 2,400,051,000
 - C. 2,400,049,000
 - D. 2,399,850,000
- 15.** Round 6.74 to the nearest tenth.
- A. 6.8
 - B. 7.0
 - C. 6.0
 - D. 6.7
- 16.** Round 15.096 to the nearest hundredth.
- A. 15.10
 - B. 15.09
 - C. 15.11
 - D. 15.96
- 17.** Round 0.997 to the nearest hundredth.
- A. 0.99
 - B. 1.00
 - C. 0.97
 - D. 1.10
- 18.** Round 243.6584 to the nearest thousandth.
- A. 243.659
 - B. 243.66
 - C. 243.658
 - D. 243.7
- 19.** Round down 3,875,240,000 to the nearest million.
- A. 3,876,000,000
 - B. 3,870,000,000
 - C. 3,800,000,000
 - D. 3,875,000,000

- 20.** The total mobile money transactions in a region of Ghana in 2025 was GH¢2,847,516,000. What is this amount to the nearest million cedis?
- A. GH¢2,848,000,000
 - B. GH¢2,847,000,000
 - C. GH¢2,800,000,000
 - D. GH¢2,850,000,000
- 21.** How many decimal places does the number 0.0405 have?
- A. 2
 - B. 4
 - C. 3
 - D. 5
- 22.** Express 3.14159 correct to 3 decimal places.
- A. 3.141
 - B. 3.14
 - C. 3.1415
 - D. 3.142
- 23.** Express 0.0568 correct to 2 significant figures.
- A. 0.056
 - B. 0.06
 - C. 0.057
 - D. 0.0568
- 24.** Express 45,932,000 correct to 3 significant figures.
- A. 45,930,000
 - B. 45,900,000
 - C. 46,000,000
 - D. 45,000,000
- 25.** How many significant figures are there in the number 6.030?
- A. 4
 - B. 3
 - C. 2
 - D. 5
- 26.** In the year 2026, the population of the world was estimated at 8,164,000,000. Express this population correct to 1 significant figure.
- A. 8,100,000,000
 - B. 8,160,000,000
 - C. 800,000,000
 - D. 8,000,000,000

- 27.** Which symbol makes the statement true? $3,205,000,000 \square 3,200,500,000$
- A. $>$
 - B. $<$
 - C. $=$
 - D. $\times$
- 28.** Arrange the following numbers in ascending order: 1,250,000,000; 1,205,000,000; 1,502,000,000; 1,052,000,000.
- A. 1,250,000,000; 1,205,000,000; 1,502,000,000; 1,052,000,000
 - B. 1,052,000,000; 1,205,000,000; 1,250,000,000; 1,502,000,000
 - C. 1,052,000,000; 1,250,000,000; 1,205,000,000; 1,502,000,000
 - D. 1,502,000,000; 1,250,000,000; 1,205,000,000; 1,052,000,000
- 29.** Which of the following numbers is the greatest?
- A. 2,099,999,999
 - B. 2,009,999,999
 - C. 2,100,000,000
 - D. 2,090,999,999
- 30.** Four mobile network companies recorded the following data usage in a year: MTN 5,107,020,000 MB; Telecel 3,910,450,000 MB; AirtelTigo 4,012,300,000 MB; Espresso 3,999,990,000 MB. Arrange the companies in order from the least to the greatest data usage.
- A. Espresso, Telecel, AirtelTigo, MTN
 - B. Telecel, AirtelTigo, Espresso, MTN
 - C. MTN, AirtelTigo, Espresso, Telecel
 - D. Telecel, Espresso, AirtelTigo, MTN
- 31.** Which symbol is used to mean “less than”?
- A. $>$
 - B. $<$
 - C. $=$
 - D. $\neq$
- 32.** What is the value of $7,000,000,000 + 40,000,000 + 90,000 + 600$?
- A. 7,400,090,600
 - B. 7,004,090,600
 - C. 7,040,090,600
 - D. 7,040,900,006
- 33.** When the number $3,2\square6,000,000$ is rounded to the nearest ten million, the result is 3,300,000,000. What digit does $\square$ represent?
- A. 6
 - B. 7
 - C. 8

D. 9

- 34.** Abena's family business recorded sales of GH¢1,075,499,996 in 2025. What is this amount to the nearest hundred thousand cedis?
- A. GH¢1,075,400,000
 - B. GH¢1,075,500,000
 - C. GH¢1,076,000,000
 - D. GH¢1,070,000,000
- 35.** Which of the following numbers becomes 5,000,000,000 when rounded to the nearest hundred million?
- A. 4,950,000,000
 - B. 4,949,000,000
 - C. 5,050,000,000
 - D. 4,850,000,000
- 36.** Express 2.0060 correct to 3 significant figures.
- A. 2.00
 - B. 2.006
 - C. 2.1
 - D. 2.01
- 37.** I am a ten-digit number. My billions digit is 4. My hundred millions digit is half of my billions digit. My ten millions digit is three times my hundred millions digit. All my other digits are zeros. What number am I?
- A. 4,620,000,000
 - B. 2,460,000,000
 - C. 4,260,000,000
 - D. 4,206,000,000
- 38.** The population of the world is about eight billion, one hundred and sixty million. Which of the following is this number written in numerals?
- A. 8,016,000,000
 - B. 8,160,000,000
 - C. 816,000,000
 - D. 8,106,000,000
- 39.** Round 6.996 correct to 2 decimal places.
- A. 7.00
 - B. 6.99
 - C. 6.96
 - D. 7.10

- 40.** Esi was asked to round 3,450,000,000 to the nearest hundred thousand and she answered 3,400,000,000. Which of the following statements is true?
- A. Esi is correct because the number ends in zeros.
 - B. Esi is correct because 5 must always be rounded down.
 - C. Esi is wrong; the correct answer is 3,450,000,000.
 - D. Esi is wrong; the correct answer is 3,500,000,000.

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) A mobile network company sent a total of 4,073,605,218 text messages in one year.

Question 2

- (a) Four companies in Ghana recorded the following sales in the 2025/26 financial year:

Question 5

- (a) The Government of Ghana budgeted GH¢12,406,385,000 for the construction of schools in the 2026/27 academic year.

Question 6

- (a) Adjoa was given the digits 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9 and asked to use each digit once to form ten-digit numbers.

Chapter 10: Ch

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. What is 3.87×100 ?
 - A. 38.7
 - B. 387
 - C. 3,870
 - D. 0.0387
2. What is $65.4 \div 10$?
 - A. 654
 - B. 0.654
 - C. 6.54
 - D. 6,540
3. What is $0.075 \times 1,000$?
 - A. 7.5
 - B. 750
 - C. 0.75
 - D. 75
4. What is $902 \div 100$?
 - A. 9.02
 - B. 90.2
 - C. 0.902
 - D. 90,200
5. Evaluate 4.6×10 .
 - A. 0.46
 - B. 460
 - C. 46
 - D. 4.6
6. What is $7/10 \times 100$?
 - A. 7
 - B. 70
 - C. 700
 - D. 0.7

- 7.** What is $\frac{3}{4} \times 1,000$?
- A. 75
 - B. 250
 - C. 3,400
 - D. 750
- 8.** The statement $17 + 26 = 26 + 17$ illustrates the
- A. associative property of addition
 - B. commutative property of addition
 - C. distributive property of multiplication
 - D. identity property of addition
- 9.** To evaluate 25×19 as $25 \times (20 - 1)$, which property of numbers is used?
- A. Commutative property
 - B. Associative property
 - C. Distributive property
 - D. Zero property
- 10.** Using mental mathematics, $12 \times 99 =$
- A. 1,188
 - B. 1,089
 - C. 1,200
 - D. 1,098
- 11.** What is $0 \times 4,387$?
- A. 4,387
 - B. 1
 - C. 43,870
 - D. 0
- 12.** The statement $36 + 0 = 36$ illustrates the
- A. commutative property
 - B. distributive property
 - C. identity property of addition
 - D. associative property
- 13.** Using mental mathematics, $48 \times 5 =$
- A. 240
 - B. 2,400
 - C. 204
 - D. 480

- 14.** Using compensation, $76 + 58 =$
- A. 124
 - B. 134
 - C. 144
 - D. 136
- 15.** Evaluate $4 \times 17 \times 25$ mentally by choosing a convenient order.
- A. 170
 - B. 1,070
 - C. 17,000
 - D. 1,700
- 16.** Kwame has GH¢50.00. He buys a book for GH¢28.50. What is his change, worked out mentally?
- A. GH¢22.50
 - B. GH¢21.05
 - C. GH¢21.50
 - D. GH¢78.50
- 17.** Amina thinks of a number. When she multiplies it by 10, she gets 85. What is the number?
- A. 850
 - B. 8.5
 - C. 0.85
 - D. 85
- 18.** The fare from Accra to Kasoa is GH¢12.00. What is the total fare for 25 passengers, worked out mentally?
- A. GH¢250.00
 - B. GH¢325.00
 - C. GH¢300.00
 - D. GH¢3,000.00
- 19.** Evaluate $3,758 + 2,467$.
- A. 6,225
 - B. 6,215
 - C. 5,225
 - D. 6,125
- 20.** Evaluate $6,004 - 2,587$.
- A. 4,417
 - B. 3,427
 - C. 3,517
 - D. 3,417

- 21.** A shop sold 4,386 bottles of water in January and 2,947 bottles in February. How many bottles were sold altogether?
- A. 7,323
 - B. 7,333
 - C. 6,333
 - D. 7,233
- 22.** Subtract 1,896 from 5,210.
- A. 4,314
 - B. 3,324
 - C. 3,686
 - D. 3,314
- 23.** Find the missing number: $2,415 + \square = 6,000$.
- A. 3,585
 - B. 3,685
 - C. 4,585
 - D. 3,595
- 24.** What is the difference between 7,031 and 2,645?
- A. 4,486
 - B. 5,386
 - C. 4,386
 - D. 4,396
- 25.** Evaluate 436×7 .
- A. 3,042
 - B. 3,052
 - C. 2,952
 - D. 3,152
- 26.** Evaluate 285×34 .
- A. 9,690
 - B. 9,590
 - C. 8,690
 - D. 9,780
- 27.** Evaluate $852 \div 6$.
- A. 124
 - B. 148
 - C. 1,420
 - D. 142

28. Evaluate $1,275 \div 15$.

- A. 75
- B. 95
- C. 85
- D. 805

29. Evaluate $6,048 \div 24$.

- A. 225
- B. 252
- C. 262
- D. 2,520

30. When 947 is divided by 8, what are the quotient and the remainder?

- A. 118 remainder 3
- B. 117 remainder 7
- C. 118 remainder 5
- D. 119 remainder 1

31. Evaluate 306×25 .

- A. 7,560
- B. 6,750
- C. 76,500
- D. 7,650

32. A farmer packs 1,440 eggs equally into boxes of 12. How many boxes does he need?

- A. 12
- B. 1,200
- C. 120
- D. 102

33. Mrs Adjei bought 14.8 kg of fish. Mrs Mensah bought 3.2 kg less than Mrs Adjei. How many kilograms of fish did they buy altogether?

- A. 26.4 kg
- B. 18.0 kg
- C. 11.6 kg
- D. 29.6 kg

34. Kofi weighs 32.5 kg. His father weighs four times as much as Kofi. What is the total weight of Kofi and his father?

- A. 130 kg
- B. 162 kg
- C. 135.5 kg
- D. 162.5 kg

- 35.** Mrs Owusu bought 52.25 m of wax print and shared it equally among her five children. How many metres did each child receive?
- A. 10.54 m
 - B. 10.45 m
 - C. 9.45 m
 - D. 10.25 m
- 36.** Efe had GH¢20.00. She bought sugar for GH¢7.35 and milk for GH¢5.80. How much money was left?
- A. GH¢6.85
 - B. GH¢7.15
 - C. GH¢13.15
 - D. GH¢6.58
- 37.** One bag of maize weighs 25.6 kg. What is the total weight of 3 such bags?
- A. 75.8 kg
 - B. 78.6 kg
 - C. 76.8 kg
 - D. 28.6 kg
- 38.** A driver bought 45.5 litres of fuel and used 18.7 litres. How much fuel is left?
- A. 27.8 litres
 - B. 26.2 litres
 - C. 64.2 litres
 - D. 26.8 litres
- 39.** Evaluate $8.64 \div 4$.
- A. 21.6
 - B. 2.16
 - C. 2.61
 - D. 0.216
- 40.** Kojo ran 3.75 km on Saturday and 4.6 km on Sunday. How far did he run altogether?
- A. 7.35 km
 - B. 8.25 km
 - C. 8.35 km
 - D. 83.5 km

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Powers of 10.

Question 2

(a) Number properties and mental mathematics.

Question 3

(a) Addition and subtraction of whole numbers.

Question 4

(a) Multiplication and division of multi-digit numbers.

Question 5

(a) Story problems involving decimals.

Question 6

(a) Creating and solving decimal story problems.

Chapter 11: Ch3

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Express $3 \times 3 \times 3 \times 3$ in index form.
 - A. 3×4
 - B. 4^3
 - C. 3^4
 - D. 12
2. The expression 2^5 means
 - A. 2×5
 - B. $2 + 2 + 2 + 2 + 2$
 - C. 5×5
 - D. $2 \times 2 \times 2 \times 2 \times 2$
3. Which of the following represents the repeated multiplication $6 \times 6 \times 6$?
 - A. 6^3
 - B. 3^6
 - C. 6×3
 - D. 18
4. Kofi arranges bottle tops into 5 heaps of 5 caps each. The total number of caps is 5×5 . How is 5×5 written in index form?
 - A. 5^3
 - B. 2^5
 - C. 5^2
 - D. 5×2
5. How many times is 7 used as a factor in the expression 7^6 ?
 - A. 7
 - B. 6
 - C. 13
 - D. 42
6. Which of the following represents $2 \times 2 \times 2 \times 2 \times 2 \times 2$?
 - A. 6^2
 - B. 12
 - C. 6×2
 - D. 2^6

- 7.** In the expression 9^4 , the base and the exponent are respectively
- A. 4 and 9
 - B. 9 and 4
 - C. 9 and 36
 - D. 13 and 9
- 8.** Express 64 as a power of 2.
- A. 2^4
 - B. 2^5
 - C. 2^6
 - D. 2^8
- 9.** Express 81 as a power of 3.
- A. 3^4
 - B. 3^3
 - C. 4^3
 - D. 9^4
- 10.** What is the value of $2^3 \times 3^2$?
- A. 36
 - B. 24
 - C. 12
 - D. 72
- 11.** Which of the following expresses 100 as a product of powers of its prime factors?
- A. 2×5^2
 - B. $2^2 \times 5$
 - C. $2^2 \times 5^2$
 - D. 2^5
- 12.** Write $5 \times 5 \times 5 \times 2 \times 2$ in index form.
- A. $5^2 \times 2^3$
 - B. $5^3 \times 2^2$
 - C. 10^5
 - D. 5×2
- 13.** Express 1,000 as a power of 10.
- A. 10^2
 - B. 10^4
 - C. 3^{10}
 - D. 10^3

- 14.** What is the value of $7^2 \times 2^3$?
- A. 98
 - B. 392
 - C. 343
 - D. 56
- 15.** Which of the following is equal to $4 \times 4 \times 4 \times 4$?
- A. 4^4
 - B. 16
 - C. 4×4
 - D. 8^2
- 16.** What is the value of 9^0 ?
- A. 0
 - B. 1
 - C. 9
 - D. 90
- 17.** Which of the following statements is true?
- A. Every natural number raised to the power 0 gives 0
 - B. $5^0 = 5$
 - C. $0^5 = 1$
 - D. Every natural number raised to the power 0 gives 1
- 18.** Evaluate $7^0 + 3^0$.
- A. 2
 - B. 10
 - C. 1
 - D. 0
- 19.** Evaluate 4×6^0 .
- A. 24
 - B. 0
 - C. 4
 - D. 10
- 20.** If $5^n = 1$, then the value of n is
- A. 1
 - B. 0
 - C. 5
 - D. 10

21. Find the value of 2^5 .

- A. 10
- B. 25
- C. 16
- D. 32

22. Find the value of 5^3 .

- A. 15
- B. 125
- C. 25
- D. 53

23. Find the value of 10^4 .

- A. 10,000
- B. 40
- C. 1,000
- D. 100,000

24. Find the value of 3^4 .

- A. 12
- B. 27
- C. 64
- D. 81

25. Find the value of 6^3 .

- A. 18
- B. 216
- C. 36
- D. 126

26. Which of the following numbers is a perfect square?

- A. 64
- B. 48
- C. 50
- D. 72

27. Evaluate $2^3 + 3^2$.

- A. 13
- B. 25
- C. 17
- D. 36

28. Evaluate $5^2 - 4^2$.

- A. 1
- B. 9
- C. 20
- D. 41

29. Evaluate $(2 \times 3)^2$.

- A. 12
- B. 18
- C. 23
- D. 36

30. Which of the following is equal to 125?

- A. 5^3
- B. 3^5
- C. 25^2
- D. 5^2

31. Evaluate $2^0 + 2^1 + 2^2 + 2^3$.

- A. 8
- B. 14
- C. 15
- D. 16

32. If $3^n = 81$, then the value of n is

- A. 3
- B. 9
- C. 27
- D. 4

33. Express 36 as a product of powers of its prime factors.

- A. 2×3^2
- B. $2^2 \times 3^2$
- C. $2^3 \times 3$
- D. 4×9

34. Find the HCF of 12 and 18.

- A. 6
- B. 2
- C. 3
- D. 36

35. Find the HCF of 24 and 36.

- A. 6
- B. 18
- C. 12
- D. 24

36. Find the HCF of 24 and 60.

- A. 4
- B. 8
- C. 6
- D. 12

37. Find the HCF of $2^3 \times 3^2$ and $2^2 \times 3^3$.

- A. 2×3
- B. $2^2 \times 3^2$
- C. $2^3 \times 3^3$
- D. $2^2 \times 3^3$

38. Find the HCF of 45 and 75.

- A. 15
- B. 5
- C. 3
- D. 225

39. Find the HCF of 16 and 48.

- A. 4
- B. 8
- C. 32
- D. 16

40. A teacher has 48 pencils and 36 rulers. She wants to make identical packs so that no item is left over. What is the greatest number of packs she can make?

- A. 6
- B. 144
- C. 12
- D. 18

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) Repeated factors and index form.

Question 2

- (a) Expressing numbers as products of powers.

Question 3

- (a) The zero exponent.

Question 4

- (a) Finding the value of numbers in index form.

Question 5

- (a) Highest Common Factor by the product of prime factors.

Question 6

- (a) Integrated problems and reasoning.

Chapter 12: Ch

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Express $\frac{1}{2}$ as a percentage.
 - A. 5%
 - B. 2%
 - C. 50%
 - D. 20%
2. Express $\frac{3}{4}$ as a decimal.
 - A. 0.34
 - B. 0.75
 - C. 7.5
 - D. 3.4
3. Which fraction in its lowest terms is equal to 20%?
 - A. $\frac{1}{5}$
 - B. $\frac{1}{20}$
 - C. $\frac{2}{5}$
 - D. $\frac{5}{2}$
4. Which of the following fractions is equivalent to 0.4?
 - A. $\frac{1}{4}$
 - B. $\frac{4}{5}$
 - C. $\frac{4}{50}$
 - D. $\frac{2}{5}$
5. Which of the following is $\frac{1}{3}$ written as a decimal?
 - A. 0.30
 - B. 3.0
 - C. 0.333...
 - D. 0.033
6. Express 40% as a decimal.
 - A. 0.4
 - B. 0.04
 - C. 4.0
 - D. 0.14

- 7.** Which of the following fractions is greater?
- A. $\frac{2}{3}$
 - B. $\frac{3}{4}$
 - C. They are equal
 - D. It cannot be determined
- 8.** Arrange the fractions $\frac{5}{6}$, $\frac{3}{4}$ and $\frac{7}{8}$ in descending order.
- A. $\frac{3}{4}$, $\frac{5}{6}$, $\frac{7}{8}$
 - B. $\frac{5}{6}$, $\frac{7}{8}$, $\frac{3}{4}$
 - C. $\frac{7}{8}$, $\frac{3}{4}$, $\frac{5}{6}$
 - D. $\frac{7}{8}$, $\frac{5}{6}$, $\frac{3}{4}$
- 9.** Which of the following decimal fractions is greater?
- A. 0.977
 - B. 0.99 and 0.977 are equal
 - C. 0.99
 - D. It cannot be determined
- 10.** Arrange the decimals 0.098, 0.985 and 0.123 in ascending order.
- A. 0.098, 0.123, 0.985
 - B. 0.123, 0.098, 0.985
 - C. 0.098, 0.985, 0.123
 - D. 0.985, 0.123, 0.098
- 11.** Which of the following is equivalent to $\frac{3}{5}$?
- A. 0.35
 - B. 30%
 - C. $\frac{5}{3}$
 - D. 0.6
- 12.** Which of the following fractions is the least?
- A. $\frac{2}{5}$
 - B. $\frac{1}{3}$
 - C. $\frac{3}{8}$
 - D. $\frac{1}{2}$
- 13.** Arrange 25%, $\frac{1}{3}$ and 0.3 in ascending order.
- A. 25%, 0.3, $\frac{1}{3}$
 - B. 0.3, 25%, $\frac{1}{3}$
 - C. $\frac{1}{3}$, 0.3, 25%
 - D. 25%, $\frac{1}{3}$, 0.3

14. Which percentage is equal to $\frac{7}{10}$?

- A. 7%
- B. 17%
- C. 70%
- D. 0.7%

15. Evaluate $\frac{1}{4} + \frac{2}{4}$.

- A. $\frac{3}{8}$
- B. $\frac{3}{4}$
- C. $\frac{2}{16}$
- D. $\frac{1}{2}$

16. Evaluate $\frac{1}{2} + \frac{1}{3}$.

- A. $\frac{2}{5}$
- B. $\frac{2}{6}$
- C. $\frac{1}{6}$
- D. $\frac{5}{6}$

17. Evaluate $\frac{2}{3} + \frac{1}{4}$.

- A. $\frac{11}{12}$
- B. $\frac{3}{7}$
- C. $\frac{3}{12}$
- D. $\frac{8}{12}$

18. Evaluate $\frac{3}{4} - \frac{1}{3}$.

- A. $\frac{2}{1}$
- B. $\frac{2}{7}$
- C. $\frac{5}{12}$
- D. $\frac{1}{6}$

19. Evaluate $2\frac{2}{5} + 1\frac{2}{3}$.

- A. $3\frac{4}{15}$
- B. $4\frac{1}{15}$
- C. $3\frac{16}{15}$
- D. $4\frac{4}{15}$

20. Evaluate $5 - 2\frac{3}{4}$.

- A. $2\frac{1}{4}$
- B. $2\frac{3}{4}$
- C. $3\frac{1}{4}$
- D. $2\frac{1}{2}$

21. Evaluate $3\frac{1}{2} - 1\frac{3}{4}$.

- A. $2\frac{1}{2}$
- B. $1\frac{1}{4}$
- C. $2\frac{1}{4}$
- D. $1\frac{3}{4}$

22. A plank is $12\frac{1}{4}$ metres long. A carpenter cuts off $3\frac{1}{3}$ metres. How long is the remaining part?

- A. $9\frac{1}{12}$ m
- B. $8\frac{1}{4}$ m
- C. $8\frac{11}{12}$ m
- D. $9\frac{1}{4}$ m

23. The Musa family hiked $2\frac{1}{2}$ km in the morning and $3\frac{3}{4}$ km in the afternoon. What was the total distance they hiked?

- A. $6\frac{3}{8}$ km
- B. $6\frac{1}{4}$ km
- C. $5\frac{3}{4}$ km
- D. $6\frac{1}{2}$ km

24. Ama drank $\frac{1}{3}$ of a bottle of water and Adjoa drank $\frac{1}{4}$ of it. What fraction of the water did they drink altogether?

- A. $\frac{7}{12}$
- B. $\frac{2}{7}$
- C. $\frac{1}{7}$
- D. $\frac{2}{12}$

25. Esi spent $\frac{1}{2}$ of her money on rice and $\frac{1}{4}$ of it on cooking oil. What fraction of her money is left?

- A. $\frac{3}{4}$
- B. $\frac{1}{2}$
- C. $\frac{1}{8}$
- D. $\frac{1}{4}$

26. Evaluate $3 \times 2\frac{2}{3}$.

- A. $6\frac{2}{3}$
- B. $7\frac{2}{3}$
- C. 8
- D. 24

27. Find $\frac{2}{3}$ of GH¢60.00.

- A. GH¢30.00
- B. GH¢40.00
- C. GH¢20.00
- D. GH¢90.00

28. What is $\frac{1}{4}$ of 100?

- A. 25
- B. 40
- C. 14
- D. 400

29. Evaluate $\frac{3}{5} \times 20$.

- A. 15
- B. 30
- C. 3
- D. 12

30. Evaluate $\frac{2}{3} \times \frac{3}{4}$.

- A. $\frac{6}{7}$
- B. $\frac{8}{9}$
- C. $\frac{1}{2}$
- D. $\frac{5}{7}$

31. In a class of 40 pupils, $\frac{5}{8}$ are girls. How many girls are in the class?

- A. 20
- B. 25
- C. 16
- D. 32

32. Evaluate 0.5×0.2 .

- A. 0.1
- B. 0.7
- C. 0.01
- D. 1.0

33. Find 25% of GH¢80.00.

- A. GH¢40.00
- B. GH¢25.00
- C. GH¢8.00
- D. GH¢20.00

34. Evaluate $10 \div \frac{1}{4}$.

- A. 2.5
- B. 10
- C. 40
- D. 25

35. Evaluate $\frac{3}{4} \div 3$.

- A. $\frac{9}{4}$
- B. $\frac{1}{4}$
- C. $\frac{1}{3}$
- D. $\frac{3}{16}$

36. Evaluate $\frac{1}{2} \div \frac{1}{3}$.

- A. $1 \frac{1}{2}$
- B. $\frac{1}{6}$
- C. $\frac{2}{3}$
- D. $\frac{1}{5}$

37. Evaluate $2 \div \frac{1}{2}$.

- A. 1
- B. $\frac{1}{4}$
- C. 8
- D. 4

38. Evaluate $\frac{5}{6} \div 5$.

- A. $\frac{25}{6}$
- B. $\frac{1}{30}$
- C. $\frac{1}{6}$
- D. $\frac{5}{6}$

39. Evaluate $0.84 \div 4$.

- A. 2.1
- B. 0.21
- C. 0.021
- D. 0.88

40. A sack of rice weighing 15 kg is packed into bags of $1 \frac{1}{2}$ kg each. How many bags are obtained?

- A. 10
- B. 15
- C. 30
- D. 5

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Benchmark fractions, percentages and decimals.

Question 2

(a) Comparing and ordering.

Question 3

(a) Addition and subtraction of fractions.

Question 4

(a) Word problems involving addition and subtraction of fractions.

Question 5

(a) Multiplying fractions and finding a fraction of a quantity.

Question 6

(a) Dividing fractions.

Chapter 13: Ch5

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. There are 60 boys and 120 girls in a school. What is the ratio of boys to girls in its simplest form?
 - A. 2 : 1
 - B. 1 : 2
 - C. 1 : 3
 - D. 180 : 60
2. Express the ratio 15 : 25 in its simplest form.
 - A. 3 : 5
 - B. 5 : 3
 - C. 1 : 2
 - D. 15 : 25
3. A box contains 8 red counters and 12 blue counters. What is the ratio of red counters to blue counters in its simplest form?
 - A. 3 : 2
 - B. 4 : 6
 - C. 1 : 2
 - D. 2 : 3
4. The ratio of mangoes to oranges in a basket is 4 : 5. If there are 20 mangoes, how many oranges are in the basket?
 - A. 16
 - B. 20
 - C. 25
 - D. 100
5. In a school the ratio of teachers to pupils is 1 : 30. If there are 4 teachers, how many pupils are there?
 - A. 34
 - B. 120
 - C. 30
 - D. 26

- 6.** The statement “4 to 9” is written in ratio notation as
- A. 4 : 9
 - B. 9 : 4
 - C. 4 : 5
 - D. 9 : 13
- 7.** Simplify the ratio 24 : 36.
- A. 12 : 18
 - B. 3 : 2
 - C. 4 : 9
 - D. 2 : 3
- 8.** Two numbers are in the ratio 5 : 3 and their sum is 40. What are the two numbers?
- A. 20 and 20
 - B. 30 and 10
 - C. 25 and 15
 - D. 15 and 25
- 9.** Two quantities are in the ratio 2 : 5. If the first quantity is 14, what is the second quantity?
- A. 35
 - B. 28
 - C. 20
 - D. 70
- 10.** In a bag, sweets and toffees are in the ratio 7 : 3. What fraction of the total number are toffees?
- A. $\frac{3}{7}$
 - B. $\frac{7}{10}$
 - C. $\frac{7}{3}$
 - D. $\frac{3}{10}$
- 11.** A recipe has a ratio of 3 cups of flour to 4 cups of sugar. How many cups of flour are used for each cup of sugar?
- A. $\frac{4}{3}$ cup
 - B. $\frac{3}{4}$ cup
 - C. 3 cups
 - D. 7 cups
- 12.** A car travels 180 km in 3 hours. What is its speed as a unit rate?
- A. 60 km/h
 - B. 183 km/h
 - C. 540 km/h
 - D. 3 km/h

- 13.** Five oranges cost GH¢12.50. What is the cost of one orange?
- A. GH¢62.50
 - B. GH¢5.00
 - C. GH¢2.50
 - D. GH¢2.05
- 14.** A machine fills 240 bottles of water in 4 minutes. How many bottles does it fill per minute?
- A. 960
 - B. 60
 - C. 236
 - D. 24
- 15.** Ama walks 6 km in 2 hours. What is her speed as a unit rate?
- A. 12 km/h
 - B. 8 km/h
 - C. 4 km/h
 - D. 3 km/h
- 16.** The ratio 7 : 4 can be written as the unit rate
- A. $\frac{7}{4}$
 - B. $\frac{4}{7}$
 - C. $\frac{11}{4}$
 - D. $\frac{7}{11}$
- 17.** Kofi types 350 words in 7 minutes. How many words does he type per minute?
- A. 343
 - B. 2,450
 - C. 50
 - D. 35
- 18.** Which of the following ratios is equivalent to 2 : 3?
- A. 3 : 2
 - B. 8 : 12
 - C. 6 : 8
 - D. 12 : 8
- 19.** Complete the equivalent ratio: $5 : 7 = 20 : \square$.
- A. 25
 - B. 30
 - C. 24
 - D. 28

- 20.** Ama and Kofi share GH¢120.00 in the ratio 2 : 3. How much does Kofi receive?
- A. GH¢72.00
 - B. GH¢48.00
 - C. GH¢60.00
 - D. GH¢240.00
- 21.** 45 sweets are shared between two children in the ratio 4 : 5. What is the smaller share?
- A. 25
 - B. 4
 - C. 20
 - D. 9
- 22.** Kafui, Adoley and Jantuah shared an amount of money in the ratio of their ages, 3 : 4 : 2. If Jantuah, the youngest, received GH¢2,400.00, what was the total amount shared?
- A. GH¢9,600.00
 - B. GH¢10,800.00
 - C. GH¢7,200.00
 - D. GH¢2,400.00
- 23.** In the sharing in Question 22, how much did Adoley, the eldest, receive?
- A. GH¢2,400.00
 - B. GH¢3,600.00
 - C. GH¢7,200.00
 - D. GH¢4,800.00
- 24.** In a table of equivalent ratios, $\frac{3}{10} = \frac{6}{x}$. What is the value of x?
- A. 20
 - B. 10
 - C. 18
 - D. 30
- 25.** The ratio of the length to the width of a rectangle is 5 : 2. If the width is 8 cm, what is the length?
- A. 10 cm
 - B. 16 cm
 - C. 20 cm
 - D. 40 cm

- 26.** GH¢72.00 is shared among three children in the ratio 1 : 2 : 3. What is the greatest share?
- A. GH¢24.00
 - B. GH¢36.00
 - C. GH¢12.00
 - D. GH¢72.00
- 27.** Find the number that makes the ratio 2 : 7 equivalent to $\square$: 35.
- A. 7
 - B. 12
 - C. 14
 - D. 10
- 28.** Paint is mixed in the ratio 2 parts red to 5 parts white. How many litres of red paint are needed to make 21 litres of the mixture?
- A. 6 litres
 - B. 15 litres
 - C. 9 litres
 - D. 2 litres
- 29.** The ratio of pens to pencils in a shop is 3 : 8. If there are 24 pencils, how many pens are there?
- A. 3
 - B. 6
 - C. 9
 - D. 64
- 30.** A table of equivalent ratios is shown: 3 → GH¢10; 6 → x; 9 → GH¢30. What is the value of x?
- A. GH¢13.00
 - B. GH¢20.00
 - C. GH¢16.00
 - D. GH¢60.00
- 31.** Using the same table of equivalent ratios as in Question 30, the entry y → GH¢40 gives the value of y as
- A. 16
 - B. 21
 - C. 10
 - D. 12
- 32.** The table shows bags of maize and their weights: 2 bags → 100 kg; 5 bags → 250 kg; 8 bags → $\square$ kg. What is the missing weight?
- A. 400 kg
 - B. 350 kg
 - C. 500 kg

D. 300 kg

33. If $x/6 = 10/3$, find the value of x .

- A. 60
- B. 5
- C. 20
- D. 30

34. The points (1, 3), (2, 6) and (3, y) lie on a straight line through the origin. What is the value of y ?

- A. 12
- B. 9
- C. 6
- D. 8

35. Which of the following points lies on the graph of the ratio 2 : 5?

- A. (5, 4)
- B. (2, 10)
- C. (10, 2)
- D. (4, 10)

36. One exercise book costs GH¢12.00. What is the cost of 5 exercise books?

- A. GH¢60.00
- B. GH¢17.00
- C. GH¢72.00
- D. GH¢48.00

37. If $6 : 15 = 10 : x$, find the value of x .

- A. 15
- B. 20
- C. 25
- D. 4

38. A tap flows at a constant rate and fills 45 litres of water in 9 minutes. How many litres will it fill in 15 minutes?

- A. 45 litres
- B. 75 litres
- C. 60 litres
- D. 90 litres

39. On a map, 1 cm represents 5 km. Two towns are 7 cm apart on the map. What is the actual distance between the towns?

- A. 12 km
- B. 5 km
- C. 7 km

D. 35 km

40. Three workers share GH¢450.00 in the ratio of the hours they worked, 2 : 3 :

4. What is the difference between the greatest and the least shares?

A. GH¢100.00

B. GH¢50.00

C. GH¢200.00

D. GH¢250.00

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) The meaning of ratio and its relationship to fractions.

Question 2

(a) Unit rates.

Question 3

(a) Tables of equivalent ratios and sharing in a ratio.

Question 4

(a) Proportional reasoning: missing values and coordinate pairs.

Question 5

(a) Mixed applications of ratio, rate and sharing.

Question 6

(a) Creating and reasoning with ratios.

Chapter 14: Ch6

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. What is the next term of the sequence 2, 5, 8, 11, ... ?
 - A. 13
 - B. 12
 - C. 14
 - D. 15
2. What is the next term of the sequence 3, 6, 12, 24, ... ?
 - A. 36
 - B. 48
 - C. 30
 - D. 96
3. The pattern 1, 4, 9, 16, ... continues with
 - A. 25
 - B. 20
 - C. 24
 - D. 36
4. A pattern of squares is made with matchsticks. Figure 1 uses 4 sticks, Figure 2 uses 7 sticks and Figure 3 uses 10 sticks. How many sticks does Figure 4 use?
 - A. 12
 - B. 16
 - C. 11
 - D. 13
5. In the matchstick pattern in Question 4, how many more sticks does each figure use than the figure before it?
 - A. 4
 - B. 3
 - C. 7
 - D. 1

- 6.** The letters A, B, C are written repeatedly in the order A, B, C, A, B, C, ... Which letter is written 8th?
- A. B
 - B. C
 - C. A
 - D. AB
- 7.** What is the next term of the sequence 100, 90, 80, 70, ... ?
- A. 65
 - B. 50
 - C. 60
 - D. 70
- 8.** A pattern of triangles is made with matchsticks: 1 triangle uses 3 sticks, 2 triangles use 5 sticks and 3 triangles use 7 sticks. How many sticks are needed for 5 triangles?
- A. 15
 - B. 10
 - C. 13
 - D. 11
- 9.** What is the next term of the sequence 1, 1, 2, 3, 5, 8, ... ?
- A. 13
 - B. 11
 - C. 12
 - D. 16
- 10.** Which rule generates the sequence 2, 5, 8, 11, ... ?
- A. Add 2 to each term
 - B. Add 3 to each term
 - C. Multiply each term by 3
 - D. Subtract 3 from each term
- 11.** "One more than twice a number" is written in symbols as
- A. $n + 2$
 - B. $2n$
 - C. n^2
 - D. $2n + 1$
- 12.** "One less than three times a number n " is written in symbols as
- A. $3n - 1$
 - B. $3 - n$
 - C. $3(n - 1)$
 - D. $n - 3$

- 13.** The rule for a sequence is $n \rightarrow 4n$. What is the 5th term of the sequence?
- A. 9
 - B. 45
 - C. 20
 - D. 40
- 14.** The rule for a sequence is $n \rightarrow n + 7$. What is the 10th term of the sequence?
- A. 70
 - B. 17
 - C. 13
 - D. 10
- 15.** Which expression means “twice a number, plus 5”?
- A. $n^2 + 5$
 - B. $2(n + 5)$
 - C. $5n + 2$
 - D. $2n + 5$
- 16.** A rule is $n \rightarrow n - 4$. What is the first term of the sequence it generates?
- A. -3
 - B. 3
 - C. -5
 - D. 5
- 17.** The expression $2n - 3$ means
- A. three times a number, minus 2
 - B. twice a number, plus 3
 - C. three less than twice a number
 - D. two less than three times a number
- 18.** A function machine gives these results: input 2 $\rightarrow$ output 7; input 3 $\rightarrow$ output 10; input 4 $\rightarrow$ output 13. What is the rule of the machine?
- A. $n + 5$
 - B. $3n + 1$
 - C. $4n - 1$
 - D. $2n + 3$
- 19.** Using the rule of the machine in Question 18, what output does the input 6 give?
- A. 18
 - B. 16
 - C. 21
 - D. 19

- 20.** What is the 6th term of the sequence 4, 9, 14, 19, ... ?
- A. 29
 - B. 24
 - C. 30
 - D. 34
- 21.** What is the next term of the pattern 2, 6, 18, 54, ... ?
- A. 108
 - B. 72
 - C. 162
 - D. 56
- 22.** What are the next two terms of the sequence 50, 45, 40, 35, ... ?
- A. 25, 20
 - B. 30, 25
 - C. 35, 30
 - D. 20, 15
- 23.** A sequence starts with 3 and each next term is found by multiplying the previous term by 2 and adding 1: 3, 7, 15, 31, ... What is the next term?
- A. 47
 - B. 62
 - C. 32
 - D. 63
- 24.** In the matchstick pattern of Question 4, the number of sticks for Figure n is given by
- A. $3n + 1$
 - B. $n + 3$
 - C. $3n$
 - D. $4n$
- 25.** What is the 8th term of the sequence whose rule is $3n + 1$?
- A. 24
 - B. 11
 - C. 25
 - D. 27
- 26.** A table of values shows x : 1, 2, 3, 4 and y : 5, 8, 11, 14. Which rule connects x and y ?
- A. $y = 2x + 3$
 - B. $y = 3x + 2$
 - C. $y = x + 4$
 - D. $y = 4x + 1$

- 27.** Using the rule $y = 3x + 2$, what is the value of y when $x = 10$?
- A. 30
 - B. 23
 - C. 132
 - D. 32
- 28.** The pattern 1, 8, 27, 64, ... continues with
- A. 125
 - B. 100
 - C. 81
 - D. 216
- 29.** What are the coordinates of the origin of the number plane?
- A. (1, 1)
 - B. (0, 1)
 - C. (0, 0)
 - D. (1, 0)
- 30.** A point is located 3 units to the right of the origin and 4 units up. What are its coordinates?
- A. (4, 3)
 - B. (3, 4)
 - C. (-3, 4)
 - D. (3, -4)
- 31.** A point has x-coordinate -2 and y-coordinate 5. The point is written as
- A. (2, 5)
 - B. (5, -2)
 - C. (-5, 2)
 - D. (-2, 5)
- 32.** The rule $x \rightarrow 2x + 1$ is applied to the input $x = -2$. What is the output?
- A. -3
 - B. -4
 - C. 3
 - D. -5
- 33.** Using the rule $x \rightarrow 2x + 1$, what is the output when $x = 3$?
- A. 6
 - B. 5
 - C. 7
 - D. 9

- 34.** For the rule $x \rightarrow 2x + 1$, which input gives an output of 5?
- A. 5
 - B. 2
 - C. 3
 - D. 1
- 35.** Which of the following points lies on the graph of $y = 2x + 1$?
- A. (1, 2)
 - B. (2, 3)
 - C. (3, 1)
 - D. (1, 3)
- 36.** The points (0, 1), (1, 3) and (2, 5) lie on a straight line. What is the next point on the line?
- A. (3, 7)
 - B. (3, 6)
 - C. (4, 7)
 - D. (2, 7)
- 37.** The point (4, y) lies on the graph of $y = 2x + 1$. What is the value of y?
- A. 8
 - B. 6
 - C. 9
 - D. 7
- 38.** The points (1, 2), (2, 4) and (3, 6) show the cost in cedis of pencils. What is the cost of 5 pencils?
- A. GH¢5.00
 - B. GH¢10.00
 - C. GH¢12.00
 - D. GH¢25.00
- 39.** Which of the following points has a negative x-coordinate and a positive y-coordinate?
- A. (2, 5)
 - B. (-2, -5)
 - C. (5, -2)
 - D. (-2, 5)
- 40.** Using the pencil-cost relation in Question 38, how many pencils can be bought for GH¢16.00?
- A. 8
 - B. 14
 - C. 16
 - D. 6

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) Extending relations and patterns.

Question 2

- (a) Describing rules in words and symbols.

Question 3

- (a) Identifying rules and making predictions.

Question 4

- (a) Tables of values and the number plane.

Question 5

- (a) Using relations to solve problems.

Question 6

- (a) Creating patterns and reasoning.

Chapter 15: Ch

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. If x represents an unknown number, "10 more than a number x " is written in symbols as
 - A. $x - 10$
 - B. $10 - x$
 - C. $x + 10$
 - D. $10x$
2. "5 less than a certain number x " is written in symbols as
 - A. $x - 5$
 - B. $5 - x$
 - C. $x + 5$
 - D. $5x$
3. "2 more than 5 times a certain number x " is written in symbols as
 - A. $5(x + 2)$
 - B. $2x + 5$
 - C. $5 + 2x$
 - D. $5x + 2$
4. Half of a certain number x is written in symbols as
 - A. $2x$
 - B. $x/2$
 - C. $x - 2$
 - D. $x + 2$
5. When 8 times a certain number x is subtracted from 5 and the result is multiplied by 2, the expression formed is
 - A. $2(5 - 8x)$
 - B. $2(8x - 5)$
 - C. $2 \times 5 - 8x$
 - D. $(5 - 8x) \div 2$

6. Afrako is 3 years older than Maako. If Maako is now x years old, Afrako's age is
- A. $x - 3$
 - B. $x + 3$
 - C. $3x$
 - D. $3 - x$
7. Agbolosu and Tetteh shared GH¢400.00. Tetteh had GH¢35.00 more than Agbolosu. If Agbolosu's share is GH¢ x , an expression for Tetteh's share is
- A. $x - 35$
 - B. $35 - x$
 - C. $400 - x$
 - D. $x + 35$
8. A woman buys a basket of oranges for GH¢ x and sells it for GH¢ y . An expression for her profit is
- A. $y - x$
 - B. $x - y$
 - C. $x + y$
 - D. xy
9. Write $x + x$ in its simplest form.
- A. x^2
 - B. xx
 - C. $2x$
 - D. $x + x$
10. Write $y + y + y + y$ in its simplest form.
- A. y^4
 - B. $4y$
 - C. $4 + y$
 - D. $3y$
11. Simplify $4x + 3x + x$.
- A. $8x$
 - B. $7x$
 - C. $8x^2$
 - D. $12x$
12. Simplify $3abc + 4abc + 2abc$.
- A. $24abc$
 - B. $a^3b^3c^3$
 - C. $9abc$
 - D. $6abc$

13. Simplify $3x - 4x - 2x$.

- A. $5x$
- B. $-5x$
- C. $3x$
- D. $-3x$

14. Simplify $5x + 4 - 9y + 3x + 2y - 7$.

- A. $8x - 7y + 3$
- B. $8x - 7y - 3$
- C. $2x - 11y - 3$
- D. $8x + 7y - 3$

15. Simplify $2p - 3q + 3p + 5q$.

- A. $5p + 2q$
- B. $5p - 2q$
- C. $5p + 8q$
- D. $6p - q$

16. Simplify $7xy + 5x - 4x + 2xy - 3$.

- A. $9xy + x + 3$
- B. $9xy - x - 3$
- C. $7xy + x - 3$
- D. $9xy + x - 3$

17. Simplify $4p \times 8p^2$.

- A. $32p^2$
- B. $32p^3$
- C. $12p^3$
- D. $32p^6$

18. Simplify $-3xy \times 5y$.

- A. $-15xy^2$
- B. $15xy^2$
- C. $-8xy^2$
- D. $-15x^2y$

19. Simplify $-2a \times 4a \times 5a$.

- A. $40a^3$
- B. $-40a^2$
- C. $-40a^3$
- D. $-11a^3$

20. Simplify $5xy^2 \times 4x^4y^3$.

- A. $9x^5y^5$
- B. $20x^5y^6$
- C. $20x^9y^6$
- D. $20x^5y^5$

21. Simplify $12x^3y^2 \div 16xy^4$.

- A. $3x^2y^2/4$
- B. $3x^2/(4y^2)$
- C. $4y^2/(3x^2)$
- D. $3/(4x^2y^2)$

22. Simplify $18x^5y^2 \div 24x^3y^2$.

- A. $3x^2/4$
- B. $3x^8/4$
- C. $4x^2/3$
- D. $3x/4$

23. Simplify $8xyz \div 16xy$.

- A. $2z$
- B. z
- C. $z/2$
- D. $xyz/2$

24. Simplify $21x^7 \div 3x^4$.

- A. $18x^3$
- B. $7x^{11}$
- C. $7x^4$
- D. $7x^3$

25. If $x = 2$ and $y = 4$, the value of $3xy \times 5y$ is

- A. 120
- B. 480
- C. 240
- D. 960

26. If $x = 2$ and $y = 4$, simplify $7xy + 5x - 4x + 2xy - 3$ and find its value.

- A. 71
- B. 75
- C. 67
- D. 79

- 27.** If $p = 3$ and $z = -1$, the value of $4p \times 8z^2$ is
- A. -96
 - B. 32
 - C. -32
 - D. 96
- 28.** If $x = 2$ and $y = 4$, the value of $5x + 4 - 9y + 3x + 2y - 7$ is
- A. 15
 - B. -15
 - C. -13
 - D. -41
- 29.** If $x = 5$, the value of $2(x + 3)$ is
- A. 13
 - B. 10
 - C. 16
 - D. 25
- 30.** If $a = 3$ and $b = -2$, the value of $a^2 + b$ is
- A. 7
 - B. 11
 - C. 1
 - D. -7
- 31.** If $m = 7$ and $n = 5$, the value of $(m + n) \div 2$ is
- A. 12
 - B. 9.5
 - C. 6
 - D. 1
- 32.** If $x = -2$ and $y = 3$, the value of $3x - 2y$ is
- A. 0
 - B. -12
 - C. 12
 - D. -6
- 33.** Simplify $3x^2y + 2xy^2 - 4x^2y - 6xy^2$.
- A. $-x^2y - 4xy^2$
 - B. $x^2y - 4xy^2$
 - C. $-x^2y + 4xy^2$
 - D. $x^2y + 4xy^2$

- 34.** Simplify $(7x + 4x - 2x) \div 3x$.
- A. $3x$
 - B. 9
 - C. $3x^2$
 - D. 3
- 35.** Simplify $(7x^2 + 2x^2) \div 3x^2$.
- A. $3x^2$
 - B. $3x$
 - C. 3
 - D. $9x^2$
- 36.** Simplify $2(3x + 4) - x$.
- A. $5x + 4$
 - B. $6x + 8$
 - C. $4x + 8$
 - D. $5x + 8$
- 37.** The statement $3(x + 4) = 3x + 12$ illustrates the
- A. commutative property
 - B. associative property
 - C. distributive property
 - D. identity property
- 38.** Simplify $4(2a - 3) + 5a$.
- A. $13a - 3$
 - B. $13a - 12$
 - C. $20a - 3$
 - D. $9a - 12$
- 39.** Simplify $10x \div 2 + 3x - x$.
- A. $7x$
 - B. $5x$
 - C. $8x$
 - D. $12x$
- 40.** Simplify $(6m + 9) \div 3$.
- A. $2m + 9$
 - B. $6m + 3$
 - C. $3m + 3$
 - D. $2m + 3$

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Forming algebraic expressions.

Question 2

(a) Adding and subtracting algebraic expressions.

Question 3

(a) Multiplying and dividing algebraic expressions.

Question 4

(a) Substituting values to evaluate expressions. In this question, take $x = 2$, $y = 4$, $p = 3$ and $z = -1$.

Question 5

(a) Using properties of operations to simplify expressions.

Question 6

(a) Using algebraic expressions in real-life situations.

Chapter 16: Ch8

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. "Think of a number, double it and subtract 7. The result is 41." If the number is x , the equation formed is
 - A. $2(x - 7) = 41$
 - B. $7 - 2x = 41$
 - C. $2x + 7 = 41$
 - D. $2x - 7 = 41$
2. The sum of the ages of two friends is 25, and the older one is 4 times the age of the younger one. If the younger one's age is x , the equation is
 - A. $x + 4 = 25$
 - B. $4x + x = 25$
 - C. $4x - x = 25$
 - D. $x + 25 = 4x$
3. Adaako and Afrakoma shared 40 oranges. Afrakoma had 6 more than Adaako. If Adaako's share is x , the equation is
 - A. $x + (x + 6) = 40$
 - B. $x + 6 = 40$
 - C. $x + (x - 6) = 40$
 - D. $6x + x = 40$
4. Which word statement matches the equation $x + x = 15$?
 - A. The difference of two equal numbers is 15
 - B. Twice a number is 30
 - C. The sum of two equal numbers is 15
 - D. A number and 15 add up to twice the number
5. Which word problem matches the equation $2x - 4 = 12$?
 - A. When 2 is taken away from 4 times a number, the result is 12
 - B. When 4 is taken away from 2 times a certain number, the result is 12
 - C. When 4 is added to 2 times a number, the result is 12
 - D. Twice a number, minus 12, gives 4

6. "Two-thirds of a certain number is 4" is written as the equation
- A. $(2/3)x = 4$
 - B. $(3/2)x = 4$
 - C. $2x + 3 = 4$
 - D. $x + 2/3 = 4$
7. Kojo had some mangoes. After giving away 5, he had 12 left. If he started with x mangoes, the equation is
- A. $x + 5 = 12$
 - B. $5 - x = 12$
 - C. $x - 5 = 12$
 - D. $12 - x = 5$
8. Ama thinks of a number, adds 8 to it and multiplies the result by 3. Her answer is 45. If the number is x , the equation is
- A. $3x + 8 = 45$
 - B. $x + 24 = 45$
 - C. $8(x + 3) = 45$
 - D. $3(x + 8) = 45$
9. The perimeter of a rectangle is 30 cm and its length is twice its width w . The equation for the perimeter is
- A. $w + 2w = 30$
 - B. $2(w + 2w) = 30$
 - C. $2w + 2 = 30$
 - D. $w \times 2w = 30$
10. The equation $x + 7 = 20$ means
- A. when 7 is added to a certain number, the result is 20
 - B. seven more than 20 is the number
 - C. a number minus 7 gives 20
 - D. when 20 is added to a number, the result is 7
11. On a balance, a bag of rice of mass x kg together with a 3 kg weight balances a 7 kg weight. The equation is
- A. $x + 7 = 3$
 - B. $x + 3 = 7$
 - C. $3x = 7$
 - D. $x - 3 = 7$
12. The solution of the equation $x + 3 = 7$ from the balance in Question 11 is
- A. 3
 - B. 10
 - C. 21
 - D. 4

- 13.** To solve $x + 3 = 7$ using the balance method, we
- A. subtract 3 from both sides
 - B. add 3 to both sides
 - C. multiply both sides by 3
 - D. divide both sides by 3
- 14.** On a balance, 2 identical balls and a 1 kg weight balance a 5 kg weight. If each ball has mass x kg, the equation is
- A. $x + 2 = 5$
 - B. $2x - 1 = 5$
 - C. $2x + 1 = 5$
 - D. $2(x + 1) = 5$
- 15.** Using the balance idea, the solution of $2x + 1 = 5$ is
- A. 3
 - B. 4
 - C. 1
 - D. 2
- 16.** To solve $3y + 4 = 2y + 8$ using the balance method, the first step of removing $2y$ from both sides gives
- A. $y + 4 = 8$
 - B. $y = 12$
 - C. $5y + 4 = 8$
 - D. $3y = 2y + 4$
- 17.** The solution of $3y + 4 = 2y + 8$ is
- A. 8
 - B. 4
 - C. 12
 - D. 2
- 18.** In a shape balance, one rhombus has mass 12 g. One pentagon and one rhombus together balance a 20 g weight. The mass of the pentagon is
- A. 32 g
 - B. 12 g
 - C. 8 g
 - D. 240 g
- 19.** In a counter model, 3 identical cups each holding x counters and 2 loose counters balance 11 counters. The equation is
- A. $3x + 2 = 11$
 - B. $x + 5 = 11$
 - C. $3(x + 2) = 11$
 - D. $3x - 2 = 11$

- 20.** When solving an equation using a balance, whatever you do to one side you must
- A. do nothing to the other side
 - B. do the opposite on the same side
 - C. undo on the same side only
 - D. do the same to the other side
- 21.** In algebra tiles, a small square unit tile represents
- A. x
 - B. x^2
 - C. 1
 - D. 0
- 22.** In algebra tiles, a rectangular tile represents
- A. x
 - B. 1
 - C. x^2
 - D. $2x$
- 23.** A tile model shows 2 rectangular tiles and 3 unit tiles on the left side, balancing 9 unit tiles on the right side. The equation modelled is
- A. $2 + 3x = 9$
 - B. $5x = 9$
 - C. $2x - 3 = 9$
 - D. $2x + 3 = 9$
- 24.** The solution of the equation modelled in Question 23 is
- A. 2
 - B. 3
 - C. 6
 - D. 4
- 25.** A zero pair in algebra tiles is
- A. two negative unit tiles
 - B. one positive unit tile only
 - C. one positive tile and one negative tile of the same type
 - D. one rectangular tile and one square tile
- 26.** To solve $x + 4 = 9$ using algebra tiles, we
- A. add 4 negative unit tiles to one side only
 - B. remove 9 unit tiles from both sides
 - C. add 4 unit tiles to both sides
 - D. remove 4 unit tiles from both sides

- 27.** A tile model shows 3 rectangular tiles on the left balancing 2 rectangular tiles and 5 unit tiles on the right. The solution of the equation is
- A. 2
 - B. 5
 - C. 3
 - D. 10
- 28.** Using tiles, the first step in solving $2x + 5 = x + 8$ is to
- A. remove 1 x-tile from each side
 - B. remove 5 unit tiles from each side
 - C. add 3 unit tiles to each side
 - D. remove all the tiles
- 29.** Solve $x + 7 = 20$.
- A. 27
 - B. 12
 - C. 13
 - D. 14
- 30.** Solve $x - 9 = 15$.
- A. 24
 - B. 6
 - C. -6
 - D. 135
- 31.** Solve $6z + 4 = 28$.
- A. 8
 - B. 4
 - C. 24
 - D. 6
- 32.** Solve $3x + 5 = 20$.
- A. 15
 - B. 8
 - C. 5
 - D. 4
- 33.** Solve $4x + 1 = 3x + 7$.
- A. 8
 - B. 5
 - C. 2
 - D. 6

34. Solve $7w + 3 = 2w + 18$.

- A. 3
- B. 5
- C. 15
- D. 4

35. Solve $20 - 3k = k + 12$.

- A. 8
- B. 2
- C. 4
- D. -2

36. Solve $2x - 7 = 41$.

- A. 17
- B. 48
- C. 24
- D. 34

37. Solve $5x - 4 = 3x + 10$.

- A. 7
- B. 3
- C. 2
- D. 5

38. In 6 years' time Kofi will be 20 years old. If his present age is x , solve $x + 6 = 20$ to find his present age.

- A. 26 years
- B. 14 years
- C. 120 years
- D. 8 years

39. Solve $4x = 48$.

- A. 44
- B. 52
- C. 192
- D. 12

40. A number multiplied by 5, then 3 added, gives 33. The number is

- A. 7.2
- B. 36
- C. 6
- D. 15

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) Translating word problems and equations.

Question 2

- (a) Modelling equations on a balance.

Question 3

- (a) Modelling equations with algebra tiles.

Question 4

- (a) Solving linear equations.

Question 5

- (a) Solving problems with linear equations.

Question 6

- (a) Creating equations and reasoning.

Chapter 17: Ch

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. An angle of 90° is called
 - A. a right angle
 - B. an acute angle
 - C. an obtuse angle
 - D. a reflex angle
2. An angle of 37° is classified as
 - A. a right angle
 - B. an obtuse angle
 - C. an acute angle
 - D. a reflex angle
3. An angle of 129° is classified as
 - A. an acute angle
 - B. an obtuse angle
 - C. a right angle
 - D. a straight angle
4. An angle of 240° is classified as
 - A. an obtuse angle
 - B. a right angle
 - C. an acute angle
 - D. a reflex angle
5. Which of the following is an obtuse angle?
 - A. 100°
 - B. 90°
 - C. 45°
 - D. 180°
6. A reflex angle can measure
 - A. 85°
 - B. 200°
 - C. 90°
 - D. 175°

- 7.** The angle turned through by the minute hand of a clock in 15 minutes is
- A. 45°
 - B. 180°
 - C. 30°
 - D. 90°
- 8.** A protractor is an instrument used for
- A. measuring angles
 - B. measuring mass
 - C. measuring length
 - D. measuring time
- 9.** The angles 30° , 90° , 150° and 270° are classified, in order, as
- A. acute, right, reflex, obtuse
 - B. right, acute, obtuse, reflex
 - C. acute, right, obtuse, reflex
 - D. acute, obtuse, right, reflex
- 10.** A straight angle measures
- A. 90°
 - B. 180°
 - C. 360°
 - D. 270°
- 11.** Which of the following is NOT an acute angle?
- A. 30°
 - B. 89°
 - C. 45°
 - D. 91°
- 12.** One complete turn measures
- A. 360°
 - B. 180°
 - C. 270°
 - D. 90°
- 13.** Two angles are complementary. If one of them is 35° , the other is
- A. 145°
 - B. 45°
 - C. 55°
 - D. 65°

- 14.** Two angles are supplementary. If one of them is 110° , the other is
- A. 80°
 - B. 70°
 - C. 90°
 - D. 290°
- 15.** Complementary angles are two angles whose sum is
- A. 180°
 - B. 360°
 - C. 45°
 - D. 90°
- 16.** Supplementary angles are two angles whose sum is
- A. 180°
 - B. 90°
 - C. 360°
 - D. 270°
- 17.** Two complementary angles are equal. Each angle measures
- A. 90°
 - B. 45°
 - C. 60°
 - D. 30°
- 18.** Two supplementary angles are in the ratio 1 : 2. The smaller angle is
- A. 90°
 - B. 45°
 - C. 30°
 - D. 60°
- 19.** An angle and an angle 20° bigger than it are complementary. The smaller angle is
- A. 35°
 - B. 55°
 - C. 25°
 - D. 70°
- 20.** The supplement of an angle of 62° is
- A. 28°
 - B. 38°
 - C. 118°
 - D. 298°

- 21.** An angle and its complement are in the ratio 2 : 3. The angle is
- A. 54°
 - B. 36°
 - C. 24°
 - D. 45°
- 22.** Three angles on a straight line are x° , $2x^\circ$ and $3x^\circ$. The value of x is
- A. 60
 - B. 45
 - C. 30
 - D. 20
- 23.** The angles x° and $(x + 40)^\circ$ are supplementary. The value of x is
- A. 70
 - B. 110
 - C. 50
 - D. 140
- 24.** Which pair of angles is complementary?
- A. 60° and 40°
 - B. 100° and 80°
 - C. 30° and 50°
 - D. 25° and 65°
- 25.** Which pair of angles is supplementary?
- A. 45° and 55°
 - B. 95° and 85°
 - C. 120° and 70°
 - D. 60° and 20°
- 26.** The supplement of a right angle is
- A. 0°
 - B. 180°
 - C. 90°
 - D. 45°
- 27.** When two straight lines intersect, the vertically opposite angles are
- A. complementary
 - B. supplementary
 - C. unequal
 - D. equal

- 28.** Two straight lines intersect. One of the angles formed is 50° . The angle vertically opposite to it is
- A. 40°
 - B. 50°
 - C. 130°
 - D. 310°
- 29.** Two straight lines intersect. One of the angles formed is 65° . Each of the two angles adjacent to it on a straight line measures
- A. 115°
 - B. 25°
 - C. 125°
 - D. 65°
- 30.** Two straight lines intersect. One angle is $3x^\circ$ and the angle vertically opposite to it is $(x + 80)^\circ$. The value of x is
- A. 20
 - B. 80
 - C. 40
 - D. 120
- 31.** Adjacent angles are angles which
- A. are always equal
 - B. always add up to 180°
 - C. are formed only by parallel lines
 - D. share a common vertex and a common arm
- 32.** Two angles lie on a straight line. One of them is 70° . The other is
- A. 110°
 - B. 20°
 - C. 290°
 - D. 90°
- 33.** Three angles at a point on one side of a straight line are 45° , 60° and x° . The value of x is
- A. 85
 - B. 105
 - C. 75
 - D. 285
- 34.** Two straight lines intersect at a point. One of the four angles formed is 3 times an adjacent angle. The smaller angle is
- A. 30°
 - B. 45°
 - C. 60°

D. 135°

- 35.** The angles around a point are 90° , 120° , 60° and x° . The value of x is
- A. 30
 - B. 60
 - C. 120
 - D. 90
- 36.** Two adjacent angles on a straight line are such that one is twice the other. The smaller angle is
- A. 60°
 - B. 120°
 - C. 45°
 - D. 90°
- 37.** The angle vertically opposite to an angle of 142° is
- A. 38°
 - B. 142°
 - C. 48°
 - D. 218°
- 38.** Two straight lines intersect and one of the angles formed is 90° . The other three angles measure
- A. 90° , 45° and 45°
 - B. 180° , 90° and 0°
 - C. 60° , 120° and 90°
 - D. 90° , 90° and 90°
- 39.** The angles 70° and 110° are best classified as a pair of
- A. complementary angles
 - B. vertically opposite angles
 - C. supplementary angles
 - D. adjacent angles
- 40.** Two angles at a point on a straight edge are x° and $(x + 20)^\circ$. The value of x is
- A. 80
 - B. 100
 - C. 70
 - D. 60

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Measuring and classifying angles.

Question 2

(a) Complementary and supplementary angles.

Question 3

(a) Angles on straight lines and at a point.

Question 4

(a) Vertically opposite angles.

Question 5

(a) Mixed angle problems.

Question 6

(a) Drawing angles and reasoning.